

# Agent Acceleration in a Stable Environment

Learning from repeated tasks to make later ones faster and cheaper

1. Can what an agent learns on earlier tasks lower the time and money of later tasks, with the cost of learning counted?
2. After how many tasks does the learning pay for itself?
3. Which parts of the agent should learning change, and should it learn from its own runs or from exploring the environment?
4. Does what it learned still work on new tasks and after the environment changes, without lowering the success rate?

# Problem: make repeated tasks faster and cheaper

An agent runs the same workflow many times with new inputs. What it learns on earlier tasks should lower the time and money of later tasks, with the cost of learning counted, without lowering the success rate.

| FORMULA                                                                                                      | MEANING                                                                                                                                                                                             | SYMBOLS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | SOURCE                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| $v(m, p) = \frac{C_m(p)}{R_m(p)}$ $T_{\text{success}}(m, p) = \frac{\mathbb{E}[T_{\text{attempt}}]}{R_m(p)}$ | <p><math>v</math> (the source’s cost-of-pass): expected cost per success on one task, retrying until it succeeds</p> <p><math>T_{\text{success}}</math> : expected time per success on one task</p> | <ul style="list-style-type: none"><li><math>m</math> : the agent, a model plus the program around it (the harness); “m” for model, as in the source</li><li><math>p</math> : one task, e.g. one expense report; “p” for problem, as in the source</li><li><math>C_m(p)</math> : expected cost (C) of one attempt, in dollars</li><li><math>R_m(p)</math> : success rate (R), the chance that one attempt succeeds</li><li><math>T_{\text{attempt}}</math> : time (T) of one attempt, in seconds</li></ul> | <p><math>v</math> : from Erol et al. (2026)</p> <p><math>T_{\text{success}}</math> : self-defined, the same form as <math>v</math></p> |
| $\bar{v}(m) = \frac{1}{n} \sum_{k=1}^n v(m, p_k)$                                                            | <p><math>\bar{v}</math> : the average over <math>n</math> tasks; <math>\bar{T}_{\text{success}}</math> and the average success rate <math>\bar{R}</math> are taken the same way</p>                 | <ul style="list-style-type: none"><li><math>n</math> : number of tasks, e.g. expense reports filed</li><li><math>p_k</math> : the <math>k</math> -th task, <math>k = 1, \dots, n</math></li></ul>                                                                                                                                                                                                                                                                                                         | Self-defined, adapted from Erol et al. (2026).                                                                                         |
| $v_n(m') = \frac{C_{\text{learn}}}{n} + \bar{v}(m')$                                                         | <p><math>v_n</math> : expected cost per task after learning, learning cost included</p>                                                                                                             | <ul style="list-style-type: none"><li><math>m'</math> : the same agent after learning, model unchanged, harness changed</li><li><math>C_{\text{learn}}</math> : cost of learning (exploring, proposing and checking changes, keeping them up to date), each counted once</li></ul>                                                                                                                                                                                                                        | Self-defined, adapted from Kapoor et al. (2025) and Cai et al. (2024).                                                                 |
| $\min (\bar{T}_{\text{success}}(m'), v_n(m'))$ <p>s. t. <math>\bar{R}(m') \geq R_0</math></p>                | <p>The goal: lower time and money per task, success held</p>                                                                                                                                        | <ul style="list-style-type: none"><li><math>R_0</math> : lowest required success rate, fixed in advance, measured on tasks not used for learning</li></ul>                                                                                                                                                                                                                                                                                                                                                | Self-defined.                                                                                                                          |
| $n^* = \frac{C_{\text{learn}}}{\bar{v}(m) - \bar{v}(m')}$                                                    | <p><math>n^*</math> : break-even, the number of tasks after which <math>v_n(m')</math> falls below <math>\bar{v}(m)</math></p>                                                                      | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Self-defined, from row 3; Kapoor et al. (2025) report a break-even point.                                                              |

Each source’s original formula, and what we changed and why: [Appendix A0 ↗](#)

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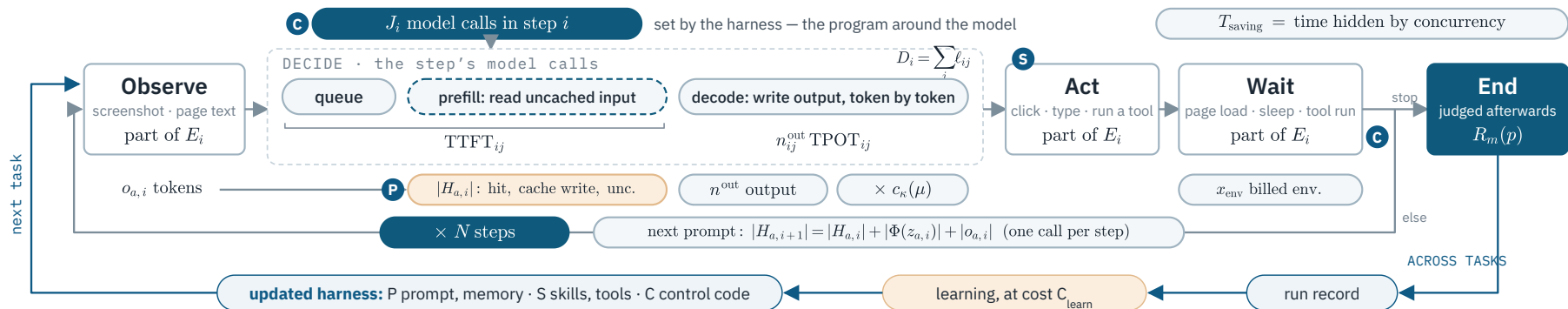
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# How an agent runs a task, and where learning changes it

One attempt is a loop of steps. Learning changes the harness to cut steps, model calls and failures; the price is a longer prompt.



● learning aims to improve it    ● always paid: learned text in the prompt; the learning bill    ● possible direction: a cached, stable prompt prefix    symbol sources: A0 ↗

P S C where the updated harness acts: prompt and memory · skills and tools · control code    may go either way: calls per step, non-model time, success

One attempt adds up to  $T_{\text{attempt}} = \sum_i (D_i + E_i) - T_{\text{saving}}$  (adapted from Hua et al., 2025; Feng et al., 2026) and  $c_m(p) = \sum_{i,j,\kappa} n_{ij}^{\kappa} c_{\kappa}(\mu_{ij}) + x_{\text{env}} c_{\text{env}}$  (adapted from Erol et al., 2026)

$N$ ,  $i$  steps in one attempt; step index (observe, decide, act, wait)

$D_i$ ,  $E_i$  model time of step  $i$ ; all its other time

$T_{\text{saving}}$  time hidden by doing things at once

$n_{ij}^{\kappa}$ ,  $\kappa$  tokens of billing class  $\kappa$ : cached, cache-written, uncached, output

$R_m(p)$  chance that one attempt of agent  $m$  on task  $p$  succeeds

$J_i$ ,  $j$  model calls in step  $i$  (e.g. plan, check, retry);  $j$  numbers them

TTFT $_{ij}$  time to the first output token: queue and prefill

$o_{a,i}$ ,  $z_{a,i}$ ,  $\Phi$  tokens the step's observation adds; the call's output; chat template

$c_{\kappa}(\mu)$ ,  $\mu$  price per token of class  $\kappa$  on the model  $\mu$  that serves the call

$c_m(p)$  dollars of one attempt; its expectation is page 1's  $C_m(p)$

$\ell_{ij}$  how long the  $j$ -th model call of step  $i$  takes

$n_{ij}^{out}$ ,  $TPOT_{ij}$  output tokens; time per output token

$|H_{a,i}|$ ,  $a$  prompt length agent  $a$  reads in step  $i$ ; which agent, when several share a task

$x_{\text{env}}$ ,  $c_{\text{env}}$  billed environment use; its price per unit

prefill reading the prompt; cached input is read faster and billed lower

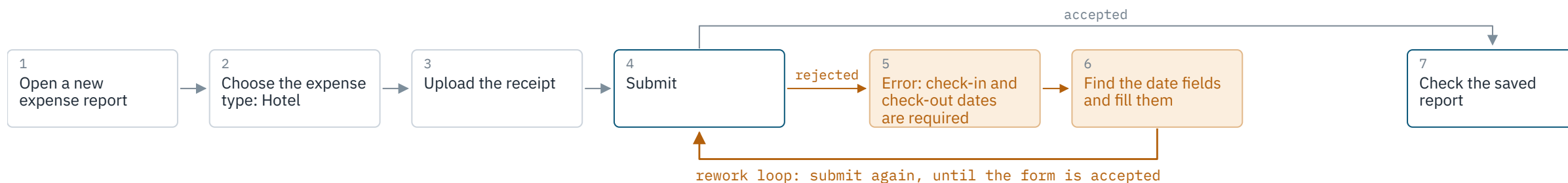
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# Example: a web agent that files expense reports

Our running example: a web agent files expense reports; each task is a new report in the same system. Not knowing one rule of the form, that a hotel expense needs check-in and check-out dates, sends the agent into a rework loop.

ONE HOTEL REPORT, BEFORE THE AGENT KNOWS THE RULE



WITHOUT LEARNING

**The loop is paid again on every hotel report**

- Its extra model calls and seconds are part of  $\bar{v}(m)$ , for each of the  $n$  reports

LEARNING THE RULE ONCE

**The rule is learned once, at cost  $C_{\text{learn}}$ ; later reports skip the loop**

- $\bar{v}(m') < \bar{v}(m)$ ; it pays off after  $n^*$  reports (page 1)

IN THIS EXAMPLE

- **Environment:** the expense system, its forms, the fields each expense type requires, the actions it allows
- **Run record:** the steps above, with the error message, and the time and cost of each step

WHAT LEARNING MAY CHANGE HERE (MODEL WEIGHTS FIXED)

- **Prompt:** an instruction read on every call: fill the check-in and check-out dates before submitting a hotel expense
- **Memory:** a note read only when relevant: Hotel → check-in and check-out dates required
- **Skill or tool:** a procedure the model calls when it needs it: instructions plus a script that fills every field a hotel expense requires
- **Control code:** the harness's own logic, run whatever the model decides: check the required fields before every submit

SCOPE

this example: expense reports on a website



any workflow repeated in one software environment



websites · desktop applications · mobile apps · tool APIs

# Five difficulties, grouped by the cost they affect (version A)

Three difficulties affect the cost of each task after learning; two affect the cost of learning itself.

| COST PER TASK AFTER LEARNING · $\bar{v}(m')$                               |                                                                                                                                                    |                                                                                                                             |                                                                                                                                        |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1 · Fewer steps do not always save time                                    | <b>WebCoach</b> (Liu et al., 2025)<br>actions 10.7 → 10.2, but 215 → 395 s per task                                                                | <b>ReasoningBank</b> (Ouyang et al., 2026)<br>steps 9.7 → 8.3, but total tokens 50,847 → 53,055 in another table            | <b>GenericAgent</b> (Liang et al., 2026)<br>fewest tokens on some tasks, yet slower than OpenClaw on five long tasks: 220.8 vs 183.1 s |
| 2 · What can be skipped is known only after a run has succeeded            | <b>EchoPath</b> (Zhao et al., 2026)<br>315.7 → 127.5 s per task, only for 159 tasks already solved; storing one took ~572k tokens                  | <b>MobileGPT</b> (Lee et al., 2024)<br>−62.5% latency, −68.8% cost on repeated tasks, after 10–15 min of exploring each app | <b>ActionEngine</b> (Zhong et al., 2026)<br>91.2% with a warmed-up site map, 73.1% without; exploring pays off after 39–101 tasks      |
| 3 · What was learned must help on new tasks and be cheap to read           | <b>Skill and memory modules</b> (Hajimiri et al., 2026)<br>with budgets roughly matched, the plain agent did best: 44.78% at 73.6K tokens per task | <b>ClawTrace</b> (Yuan et al., 2026)<br>moved to SkillsBench, median cost per task \$0.143 vs \$0.144: no saving            | <b>SEDM</b> (Xu et al., 2025)<br>its memory raised the prompt on FEVER to 2.47M tokens, from 1.65M without memory                      |
| COST OF LEARNING · $C_{\text{learn}}$                                      |                                                                                                                                                    |                                                                                                                             |                                                                                                                                        |
| 4 · Finding the right change from a failure is costly, and not guaranteed  | <b>HarnessFix</b> (Chen et al., 2026)<br>completion +6.3 to +18.4 points, for 37.2M tokens of repairs on AppWorld                                  | <b>ESPO</b> (L. Liu et al., 2026)<br>about 3.4M tokens and 2 h 13 min to optimise one prompt (PUPA)                         | <b>GEPA</b> (Agrawal et al., 2026)<br>1,839–7,051 rollouts to optimise the prompts of one benchmark                                    |
| 5 · Learning, with its testing, can cost more than the mistakes it removes | <b>SICA</b> (Robeyns et al., 2025)<br>about \$7,000 for one full run, to save \$0.21 per task (calc.)                                              | <b>ADAS</b> (Hu et al., 2025)<br>about \$500 for one search and its evaluation on ARC                                       | <b>GEA</b> (Weng et al., 2026)<br>about \$13,000 for one full evolution (author estimate)                                              |

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# Five difficulties, explained (version B)

COST PER TASK AFTER LEARNING ·  $\bar{v}(m')$

### 1 · Fewer steps do not always save time

- A learned hint, or a second model that coaches, adds reading and thinking to every step
- Steps differ in cost: skipping two clicks saves little if each remaining step waits longer
- So judge a report by its seconds and dollars, not by its step count

Works: WebCoach (Liu et al., 2025) · ReasoningBank (Ouyang et al., 2026) · GenericAgent (Liang et al., 2026)

### 2 · What can be skipped is known only after a run has succeeded

- Before the first hotel report goes through, the agent cannot tell the rework from the necessary steps
- Savings start once a successful run is stored, and storing it costs a full first pass
- A kind of report never seen before still starts from scratch

Works: EchoPath (Zhao et al., 2026) · MobileGPT (Lee et al., 2024) · ActionEngine (Zhong et al., 2026)

### 3 · What was learned must help on new tasks and be cheap to read

- A note learned on hotel reports may not help with meal or travel reports
- Every note in the prompt is read again on every call; too many notes cost more than they save
- A misleading note can even lower the success rate

Works: skill and memory modules (Hajimiri et al., 2026) · ClawTrace (Yuan et al., 2026) · SEDM (Xu et al., 2025)

COST OF LEARNING ·  $C_{\text{learn}}$

### 4 · Finding the right change from a failure is costly, and not guaranteed

- The run record shows that a report was rejected, not whether the prompt, a skill or a check should change
- Diagnosing, writing a change and re-running tasks all cost model calls
- The change found can be wrong, fix only some failures, or break other tasks

Works: HarnessFix (Chen et al., 2026) · ESPO (L. Liu et al., 2026) · DRAFT (Qu et al., 2025)

### 5 · Learning, with its testing, can cost more than the mistakes it removes

- To trust a change, it must be re-run on many reports, and those runs cost money
- Without learning, the agent simply pays for its rework on each report
- Learning pays off only after enough reports: the break-even  $n^*$  of page 1

Works: SICA (Robeyns et al., 2025) · ADAS (Hu et al., 2025) · GEA (Weng et al., 2026)

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# Five difficulties (version C: diagrams)

COST PER TASK AFTER LEARNING ·  $\bar{v}(m')$

1 · Fewer steps do not always save time

BEFORE

AFTER

one step

extra reading and thinking in each step

length = seconds per task · per step 20.1 → 38.7 s (calc.)

Each step got longer: judge a report by its seconds, not its steps

Drawn from WebCoach (Liu et al., 2025). Also: ReasoningBank (Ouyang et al., 2026), fewer steps, more tokens

2 · Whether a check is needed is not known in advance

Choose type: Hotel

check the form

fill dates

submit

✓

schematic

one step more, always right

skip: as last time

submit

✓ rule still holds

✗ it changed: rework

Skipping a check saves a step only while the rule still holds

Seen in: DRAFT (Qu et al., 2025), learns tool conditions by trying · SKILL.nb (Hattami et al., 2026), checks before reuse

3 · A trade-off between generalization and cost per use

generalizes ↑

cheap and general?

note in every prompt

note in memory

skill: steps + script

replayed script

schematic

cost per use →

Seen in: Metis (Dai et al., 2026), keeps both text and code · ActionEngine (Zhong et al., 2026), turns a site map into programs

COST OF LEARNING ·  $C_{\text{learn}}$

4 · Finding the cause behind an error message

Error: over the meal limit

possible causes, of different kinds, checked one by one

reasoning wrong type?

rule limit changed?

interaction wrong field?

tool amount misread?

...

schematic

the cause may be found late, or not at all

cost of checking so far →

Each possible cause costs a check; the right one may come late, or never

Seen in: HarnessFix (Chen et al., 2026), finds the faulty step · ESPO (L. Liu et al., 2026), sorts errors first

5 · Learning and testing can cost more than they save

total cost ↑

learning and testing are paid first

break-even  $n^*$

break-even, more testing

repair after the site changes

no learning more testing with learning

schematic

number of reports filed →

Learning pays off only if enough reports share its cost

Seen in: AI Agents That Matter (Kapoor et al., 2025), late break-even · SICA (Robeyns et al., 2025), large bill, small saving

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# Direction 1 · Learning from the agent’s own runs: 93 works in four classes

Of 93 works that change what sits outside the model, 21 measured the seconds or dollars of running tasks, 19 what learning cost, and 28 tested on unseen tasks.

PER TASK  $v_n(m') =$

$\frac{C_{\text{learn}}}{n}$

↑

+

$\bar{v}(m')$

↓

s. t.  $\bar{R}(m') \geq R_0$

| CLASS                        | WORKS | WHAT CHANGES                                                | REPRESENTATIVE WORK: ONE NUMBER AND ITS CONDITIONS                                                                                                                  | TIME OR MONEY OF RUNS<br>YES · PARTLY | LEARNING COST<br>YES · PARTLY | UNSEEN TASKS<br>YES · PARTLY |
|------------------------------|-------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------|------------------------------|
| Prompts and context<br>A02 ↗ | 15    | the prompt, instructions or context read on the next task   | ACE: offline adaptation lifts AppWorld from 42.4 to 59.4 (with labels), and takes 9,517 s against GEPA’s 53,898 s — DeepSeek-V3.1 (Q. Zhang et al., 2026)           | 1 · 2                                 | 5 · 5                         | 7 · 6                        |
| Experience memory<br>A03 ↗   | 26    | retrievable experience: successes, failures, workflows      | AWM: 35.5% vs 23.5% success on WebArena (the baseline also reads HTML), 5.9 vs 7.9 steps against an accessibility-tree BrowserGym — GPT-4-0613 (Wang et al., 2025a) | 5 · 15                                | 5 · 4                         | 11 · 14                      |
| Skills and tools<br>A04 ↗    | 34    | callable functions, scripts, skill documents                | AXIS: 29.9 s vs 59.5 s and \$0.2 vs \$0.4 per task, 84% vs 52% success, against UFO on 50 Microsoft Word tasks; learning cost not reported (Lu et al., 2025)        | 9 · 14                                | 6 · 11                        | 8 · 23                       |
| Whole harness<br>A05 ↗       | 30    | the outer program: tools, control loop, the improver itself | SICA: \$1.91 → \$1.70 and 130.2 → 114.5 s per task, for a whole run of about \$7,000; the benchmark that picked the agent also scored it (Robeyns et al., 2025)     | 8 · 4                                 | 7 · 16                        | 5 · 23                       |
| All, each work once          | 93    |                                                             |                                                                                                                                                                     | 21 · 28                               | 19 · 33                       | 28 · 57                      |

Yes · partly: our initial judgment from the per-work table (calc.; Appendix A1). A work in two classes counts in both.

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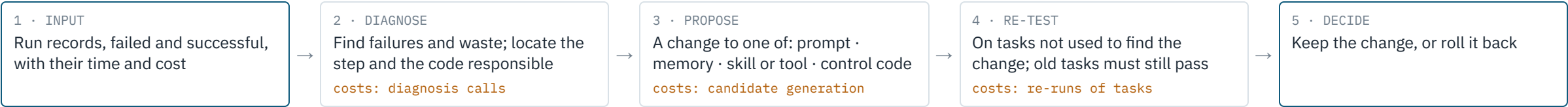
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# How a run record becomes a change, and what the change costs

Every step after reading the run records costs model calls, and these costs are rarely reported next to the savings.

PER TASK  $v_n(m') = \frac{C_{\text{learn}}}{n} \uparrow + \bar{v}(m') \downarrow \quad \text{s. t.} \quad \bar{R}(m') \geq R_0$

targeted in exchange unchanged



IT WORKS

Changes can be found and made automatically, with gains on the final tasks

- Highest success in five of six settings and 76.0–91.8% fewer model calls at deployment — Growing Harness, BrowseComp-Plus and WebArena-Verified, three models, 50 final tasks per setting, three runs (L. Li et al., 2026)

LEARNING COST LEFT OUT ·  $C_{\text{learn}}$

Savings at deployment are usually reported without the learning that produced them

- 97.4K vs 112.6K tokens and 11.25 vs 14.55 turns per AppWorld task once the memory is frozen; the memory manager’s own calls are not included — Metis (Dai et al., 2026)

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# Direction 2 · Learning the environment: 29 works keep four kinds of knowledge

Of 29 works that keep knowledge about the environment, 14 measured the seconds or dollars of running tasks, 5 what exploring cost, and 8 tested on unseen tasks.

PER TASK  $v_n(m') = \frac{C_{\text{learn}}}{n} \uparrow + \bar{v}(m') \downarrow$  s. t.  $\bar{R}(m') \geq R_0$

| CLASS                                          | WORKS | WHAT CHANGES                                            | REPRESENTATIVE WORK: ONE NUMBER AND ITS CONDITIONS                                                                                                          | TIME OR<br>MONEY OF<br>RUNS<br>YES ·<br>PARTLY | LEARNING<br>COST<br>YES ·<br>PARTLY | UNSEEN<br>TASKS<br>YES ·<br>PARTLY |
|------------------------------------------------|-------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------|------------------------------------|
| Predicting what an action does<br>A06 ↗        | 3     | what the page will look like after an action            | WMA: 140.3 s vs 748.3 s and \$0.4 vs \$2.7 per instruction against tree search, 16.6% vs 19.2% success — WebArena, GPT-4o (Chae et al., 2025)               | 1 · 1                                          | 1 · 2                               | 1 · 2                              |
| Descriptions, facts and preconditions<br>A07 ↗ | 13    | tool parameters and errors; facts and traps of the site | DRAFT: correct tool paths 88.00 vs 71.00 for ReAct on RestBench TMDB, GPT-4o; time and cost not reported (Qu et al., 2025)                                  | 6 · 4                                          | 2 · 3                               | 3 · 8                              |
| Page and place structure<br>A08 ↗              | 7     | a map of pages, actions and the paths between them      | MobileGPT: −62.5% latency and −68.8% cost on repeated tasks with human-repaired paths; exploring took 10–15 min per app, \$10.78 in all (Lee et al., 2024)  | 5 · 2                                          | 1 · 3                               | 1 · 6                              |
| Exploring and practising<br>A09 ↗              | 9     | self-set practice tasks, turned into tools              | WALT: 50.1% and 52.9% success on WebArena and VisualWebArena; exploring and checking cost \$1.67 per tool, repaid after about 14 uses (Prabhu et al., 2026) | 3 · 2                                          | 1 · 3                               | 4 · 4                              |
| All, each work once                            | 29    |                                                         |                                                                                                                                                             | 14 · 8                                         | 5 · 10                              | 8 · 18                             |

Yes · partly: our initial judgment from the per-work table (calc.; Appendix A1). A work in two classes counts in both.

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# Exploring a site, keeping a checked rule, and using it on the next task

The agent probes the form, writes down what it found together with where it applies and the evidence for it, and reads that note on the next task, so the rework loop of the expense example does not happen.

1 · EXPLORE THE FORM

Change the expense type from Meal to Hotel. Two new required fields appear: check-in and check-out date. Submitting without them returns an error.

2 · KEEP A CHECKED RULE

site-guide.md

scope: new expense form  
IF type = Hotel → check-in and check-out dates required  
evidence: 2 probes, 1 error message  
re-check: when the form changes

3 · USE IT ON THE NEXT TASK

New hotel expense + the live page + the saved note → fill every required field → submit once → check the saved report.



REAL CALLS CORRECT THE DESCRIPTION

**Trying an action and reading its result can fix what a description gets wrong**

- DRAFT learned from a real call that a tool needs a valid person\_id, and wrote it into the tool’s documentation (Qu et al., 2025)

ONE OBSERVATION IS NOT A RULE

**A rule written from one observation can be wrong, and a local check can miss it**

- DRAFT generalised from one error message; SkillWeaver’s local checks passed broken functions (Qu et al., 2025; Zheng et al., 2025)

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# Across both directions, what learning costs is rarely counted

The learning bill is rarely itemised, proving that a change is correct is the weak link, and a saving measured as fewer steps, or in one setting, does not guarantee less time or money elsewhere.

LEARNING COST ·  $C_{\text{learn}}$

Few works count both what learning costs and what it saves

- 5 of 102 works measured both the seconds or dollars of running tasks and a learning cost: GPTSwarm, MobileGPT, OpenSkill, EchoPath and Unbrowse (initial judgment, calc.) (Zhuge et al., 2024; Lee et al., 2024; Yan et al., 2026; Zhao et al., 2026; Tham et al., 2026)

CORRECTNESS ·  $\bar{R}(m') \geq R_0$

Showing that a change fixed the target without breaking what worked is the open problem

- In AgentDevel’s ablations, the highest mean score and the fewest regressions on old tasks may have to be traded against each other (Zhang, 2026)

FEWER STEPS ·  $\bar{v}(m')$

Fewer steps is not the same as less time or money

- Steps 9.7 → 8.3 on WebArena, while total tokens rose 50,847.4 → 53,054.5 in another table whose base set-up is not stated — ReasoningBank (Ouyang et al., 2026)

TRANSFER ·  $\bar{v}(m')$

A saving in one setting need not carry over

- Median cost per task \$0.143 vs \$0.144 after moving to SkillsBench, no fall — ClawTrace, OpenClaw with gpt-5.4 (Yuan et al., 2026)

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# The closest works save time or money on repeated tasks; none counts the whole loop

In the works we checked, no single result covers all of these at once: any task stream, no training, automatic location of failures and waste, protection of old abilities, every learning and maintenance cost counted, and a net saving over time.

| WORK                                           | WHAT IT LEARNS                                      | WHAT IT SHOWS: ONE NUMBER AND ITS CONDITIONS                                                                                                           | WHAT IS LEFT OUT                                                                                         |
|------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>ActionEngine</b><br>(Zhong et al., 2026)    | a state-machine map of the site, with templates     | 91.2% success (73.1% without the warmed-up map); 27 s vs 87 s and \$0.05 vs \$0.40 per task against Claude Code — WebArena, 655 tasks, Claude Opus 4.6 | per-task numbers exclude exploration and warm-up; break-even after 39–101 tasks still omits warm-up      |
| <b>WALT</b><br>(Prabhu et al., 2026)           | tools written from exploring each site              | \$1.67 per tool to explore and check it, repaid after about 14 uses                                                                                    | repairing tools after deployment is left to future work                                                  |
| <b>SpeedRunner</b><br>(Huang et al., 2026)     | programmatic skills                                 | BabyAI usage about one eighth, success from about 67% to near-perfect — gpt-5.4-mini, 200 online tasks, 30 held out, three seeds                       | output tokens and dollars disagree in one figure; total learning cost [TBD: appendix A.3, to be re-read] |
| <b>HarnessFix</b><br>(Chen et al., 2026)       | repairs to the outer program from failed runs       | completion +6.3 to +18.4 points, mean of three runs                                                                                                    | time and cost of running tasks not reported                                                              |
| <b>Growing Harness</b><br>(L. Li et al., 2026) | code changes from function-level failures           | inference cost at deployment −74.4% to −98.6%                                                                                                          | evaluator and offline optimiser not counted; break-even not measured                                     |
| <b>Metis</b><br>(Dai et al., 2026)             | text and code memory, frozen before the final tasks | 97.4K vs 112.6K tokens per AppWorld task                                                                                                               | memory manager not counted; apps never seen before not tested                                            |

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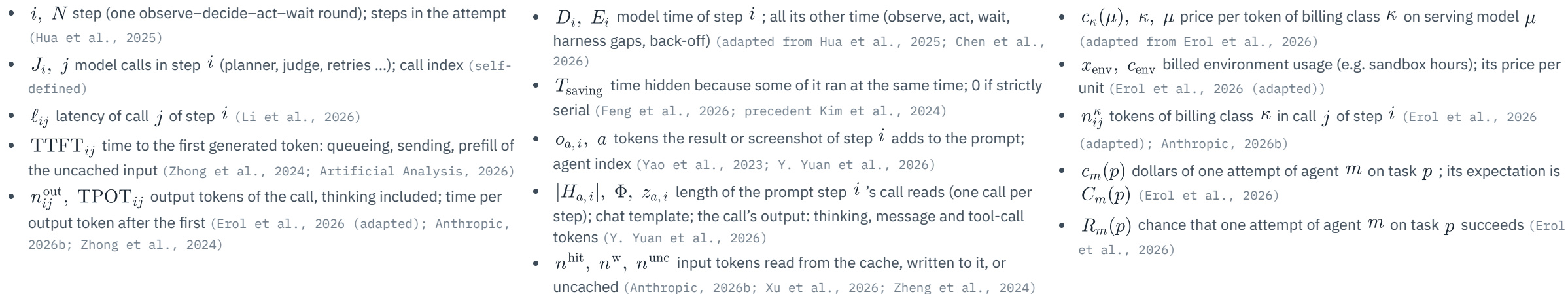
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# A0 · Page 1’s formulas: the original, ours, and why we changed it

| OUR FORMULA                                                                                                       | ORIGINAL FORMULA                                                                                                                                                            | ORIGINAL DEFINITIONS                                                                                                                                                                                                                                                                                                                                                                    | WHAT WE CHANGED, AND WHY                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $v(m, p) = \frac{C_m(p)}{R_m(p)}$ <p>from the paper</p>                                                           | $v(m, p) = \frac{C_m(p)}{R_m(p)} \text{ (Eq. 2)}$ <p>(Erol et al., 2026)</p>                                                                                                | <ul style="list-style-type: none"><li><math>m</math> : a model; <math>p</math> : a problem</li><li><math>C_m(p)</math> : expected dollars of one attempt, input and output tokens times their prices (Eq. 13)</li><li><math>R_m(p)</math> : chance that one attempt is correct; <math>1/R_m(p)</math> attempts are expected until the first correct one, attempts independent</li></ul> | <ul style="list-style-type: none"><li><math>m</math> becomes the agent: the model plus its harness</li><li>why: learning changes the harness, not the model</li></ul>                                                                                                                                                       |
| $T_{\text{success}}(m, p) = \frac{\mathbb{E}[T_{\text{attempt}}]}{R_m(p)}$ <p>self-defined, adapted</p>           | no time formula<br>(Erol et al., 2026)                                                                                                                                      | <ul style="list-style-type: none"><li>units per attempt other than dollars, such as time or latency, may matter more (App. D.1)</li></ul>                                                                                                                                                                                                                                               | <ul style="list-style-type: none"><li>the same form, in seconds</li><li>why: time and money are reported side by side</li><li>assumes attempts run one after another</li></ul>                                                                                                                                              |
| $\bar{v}(m) = \frac{1}{n} \sum_{k=1}^n v(m, p_k)$ <p>self-defined, adapted</p>                                    | $V_{p \sim D}(\mathcal{M}) \approx \mathbb{E}_{p \sim P}[V_p(\mathcal{M})] \text{ (Eq. 8)}$ <p>(Erol et al., 2026)</p>                                                      | <ul style="list-style-type: none"><li><math>D</math> : a dataset of problems; <math>P</math> : its empirical distribution, mass <math>1/n</math> on each problem</li><li><math>V_p(\mathcal{M})</math> : the lowest cost per success on <math>p</math> among the available models</li></ul>                                                                                             | <ul style="list-style-type: none"><li>the same average over <math>n</math> tasks, applied to one agent’s <math>v</math> instead of the cheapest model’s; <math>\bar{T}_{\text{success}}</math> and <math>\bar{R}</math> the same way</li><li>why: we compare one agent before and after learning</li></ul>                  |
| $v_n(m') = \frac{C_{\text{learn}}}{n} + \bar{v}(m')$ <p>self-defined, adapted</p>                                 | total cost = fixed cost + variable cost (in words)<br>(Kapoor et al., 2025)<br><br>$O(nc + C)$ against $O(nC)$ (Table 2)<br>(Cai et al., 2024)                              | <ul style="list-style-type: none"><li>fixed cost: paid once, to optimise the agent’s design (prompt, hyperparameters)</li><li>variable cost: paid on every run, set by its input and output tokens</li><li><math>n</math> : task instances; <math>C</math> : one call of the large tool-making model; <math>c</math> : one call of the small tool-using model</li></ul>                 | <ul style="list-style-type: none"><li>fixed cost ÷ <math>n</math> : a cost per task</li><li>variable part = the average cost per success <math>\bar{v}(m')</math> , so failed attempts are paid for</li><li>fixed cost widened to exploring, proposing, checking and upkeep — why: our learning does all of these</li></ul> |
| $\min (\bar{T}_{\text{success}}(m'), v_n(m'))$ <p>s. t. <math>\bar{R}(m') \geq R_0</math></p> <p>self-defined</p> | $V_p(\mathcal{M}) = \min_{m \in \mathcal{M}} v(m, p) \text{ (Eq. 3)}$ <p>(Erol et al., 2026)</p><br>accuracy and cost optimised jointly (in words)<br>(Kapoor et al., 2025) | <ul style="list-style-type: none"><li><math>\mathcal{M}</math> : the available models</li><li><math>V_p(\mathcal{M})</math> : the lowest cost per success on <math>p</math> among them</li></ul>                                                                                                                                                                                        | <ul style="list-style-type: none"><li>one number becomes a pair, seconds and dollars, lowered together — why: no fixed exchange rate between them</li><li>learning cost included</li><li>success floor <math>R_0</math> , fixed in advance — why: an agent must not look cheaper by succeeding less</li></ul>               |
| $n^* = \frac{C_{\text{learn}}}{\bar{v}(m) - \bar{v}(m')}$ <p>self-defined</p>                                     | no formula (in words)<br>(Kapoor et al., 2025)                                                                                                                              | <ul style="list-style-type: none"><li>the jointly optimised agent is cheaper in total than the default one after about 1,350 HotPotQA tasks</li></ul>                                                                                                                                                                                                                                   | <ul style="list-style-type: none"><li>written as a formula by setting <math>v_n(m') = \bar{v}(m)</math> — why: computable from measured costs</li><li>assumes the saving per task stays the same</li></ul>                                                                                                                  |



# A1 · How the three columns were judged, and the totals per class

| CLASS                                                       | WORKS | TIME OR MONEY OF RUNS (YES · PARTLY) | LEARNING COST (YES · PARTLY) | UNSEEN TASKS (YES · PARTLY) |
|-------------------------------------------------------------|-------|--------------------------------------|------------------------------|-----------------------------|
| Prompts and context                                         | 15    | 1 · 2                                | 5 · 5                        | 7 · 6                       |
| Experience memory                                           | 26    | 5 · 15                               | 5 · 4                        | 11 · 14                     |
| Skills and tools                                            | 34    | 9 · 14                               | 6 · 11                       | 8 · 23                      |
| Whole harness                                               | 30    | 8 · 4                                | 7 · 16                       | 5 · 23                      |
| Predicting what an action does                              | 3     | 1 · 1                                | 1 · 2                        | 1 · 2                       |
| Descriptions, facts and preconditions                       | 13    | 6 · 4                                | 2 · 3                        | 3 · 8                       |
| Page and place structure                                    | 7     | 5 · 2                                | 1 · 3                        | 1 · 6                       |
| Exploring and practising                                    | 9     | 3 · 2                                | 1 · 3                        | 4 · 4                       |
| Environments that check the final state (no learning)       | 3     | 1 · 2                                | 0 · 0                        | 3 · 0                       |
| Direction 1, each work once                                 | 93    | 21 · 28                              | 19 · 33                      | 28 · 57                     |
| Direction 2, each work once (final-state checkers excluded) | 29    | 14 · 8                               | 5 · 10                       | 8 · 18                      |

**Time or money of runs** — yes: measured seconds or dollars of executing tasks with what was learned; partly: tokens, steps, calls or an author’s estimate only. **Learning cost** — yes: a measured amount for learning, building or maintaining (tokens, dollars, time, compute or rollouts); partly: an estimate, iteration counts only, or a part knowingly left out. **Unseen tasks** — yes: tested on tasks that took no part in learning or selection; partly: an online stream that learns as it goes, the same workflow with new parameters, or an unstated split. These are our initial judgments from the cells of the per-work table (calc.), not re-read from the full texts; the representative works on pages 4 and 6 are to be checked one by one before the deck is final.

# A02 · Prompts and context (1/2)

| WORK                                                      | VENUE        | WHAT IT CHANGES                                                           | WHERE IT WAS TESTED                                                                                     | EFFECT, WITH CONDITIONS                                                                                                                                                       | TIME, MONEY, LEARNING COST                                                                                                                             | MARKS |
|-----------------------------------------------------------|--------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Reflexion</b><br>(Shinn et al., 2023)                  | NeurIPS 2023 | Text reflection added to next attempt on same task; model not updated     | ALFWorld, QA, code; ALFWorld retried in same environment                                                | ReAct GPT-3 prompt setting; ALFWorld cumulative 130 of 134 tasks; WebShop no improvement by four trials                                                                       | Report 12 consecutive trials; tokens, cost, time not reported; trial count ≠ steps                                                                     | ○ ○ ○ |
| <b>Self-Refine</b><br>(Madaan et al., 2023)               | NeurIPS 2023 | Current output and feedback history for next round                        | Text rewriting, code, math; per-question testing with answer changes                                    | GPT-4 code optimization 27.3% → 36.0%; GPT-3.5 math 64.1 → 64.1; compared with same model single generation                                                                   | Iteration limit 4; tokens, cost, time not reported                                                                                                     | ○ ○ ○ |
| <b>TextGrad</b><br>(Yuksekgonul et al., 2025)             | Nature 2025  | Text variables; cross-question prompts or current-question solutions      | Math, QA, code; prompt experiment split train/validation/test                                           | gpt-3.5-turbo-0125 execution, gpt-4o feedback; Object Counting 77.8 → 91.9 vs DSPy 84.9; result table does not clearly mark split assignment                                  | Prompt optimization batch size 3, 12 iterations; 1+1+1 model calls per code round; time and cost not measured                                          | ○ ● ● |
| <b>MIPRO</b><br>(Opsahl-Ong et al., 2024)                 | EMNLP 2024   | Module instructions and examples                                          | Multi-hop QA, evidence retrieval, classification; final held-out test                                   | Llama-3-8B HotPotQA test exact match 46.4 vs unoptimized 36.1, mean of 5 runs; Iris 88.6 vs Bootstrap RS 94.1                                                                 | Only trial budget evaluated; actual measured tokens, total calls, time, cost not reported                                                              | ○ ● ● |
| <b>ProTeGi</b><br>(Pryzant et al., 2023)                  | EMNLP 2023   | Prompt text                                                               | Classification task; train, dev, test separated, but final candidate selection by test F1 not clarified | Default gpt-3.5-turbo; Jailbreak test F1 when matching total API query budget: single selection after search 0.80, greedy 0.82, multi-candidate search 0.85; ~3 steps to peak | Author reported optimization sometimes past 1 hour, not per-setting timed table; deployment cost and time not measured                                 | ○ ● ● |
| <b>Semantic Backpropagation</b><br>(W. Wang et al., 2024) | arXiv        | Fixed text instructions in computational graph                            | GSM8K, BBH, BigCodeBench; main comparison uses held-out test                                            | gpt-4o-mini execution, gpt-4-turbo feedback update: GSM8K 93.2 vs TextGrad 78.2; BigCodeBench 27.8 vs 27.6; run count not reported                                            | LIAR forward input tokens 273,300 vs removed neighbor info 289,850; optimization process metric, time and cost not measured                            | ○ ● ● |
| <b>Trace / OptoPrime</b><br>(Cheng et al., 2024)          | NeurIPS 2024 | Prompts, function code or text-representable parameters                   | BBH held-out questions, controlled held-out initial conditions, new chess positions                     | BBH with reasoning template overall 78.6 vs DSPy prompt optimization 71.6; optimizer GPT-4-0125-Preview, in-program model not clearly stated                                  | Word Sorting vs TextGrad: total experiment time 82.5 vs 211.1 min, including optimization, validation and testing; not per-question deployment latency | ○ ● ● |
| <b>PROMST</b><br>(Chen et al., 2024)                      | EMNLP 2024   | Action prompts; separately train Longformer scorer, not all models frozen | Web, text environments, planning; main performance independence on held-out not clarified               | GPT-4 average progress score 0.69 vs PromptAgent 0.61; WebArena 0.62 vs APO 0.64; progress score ≠ full-task success rate                                                     | Time, cost, tokens or total calls not measured; some tasks with higher score need more steps, collisions also increase                                 | ○ ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A02 · Prompts and context (2/2)

| WORK                                               | VENUE                                         | WHAT IT CHANGES                                                                          | WHERE IT WAS TESTED                                                                                | EFFECT, WITH CONDITIONS                                                                                                                          | TIME, MONEY, LEARNING COST                                                                                                                                     | MARKS |
|----------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Dynamic Cheatsheet</b><br>(Suzgun et al., 2026) | EACL 2026                                     | Online-maintained text, strategies and code snippets                                     | Math, science QA; accumulated in single test stream                                                | GPT-4o Game of 24: baseline 10%, fixed-prompt zero memory 19%, retrieved organized memory 99%; Claude 3.5 Sonnet baseline 12% → 14%              | AIME 2024 Claude: 370 / 494 / 1035 / 1831 tokens; memory version higher; token types and management overhead range unclear; time and cost not measured         | ● ○ ● |
| <b>GPTSwarm</b><br>(Zhuge et al., 2024)            | ICML 2024 main conference                     | Node prompts, inter-agent connection probability; not LLM weights                        | GPT-3.5/GPT-4; MMLU, Crosswords, HumanEval                                                         | Crosswords same 20 questions optimization evaluation 0.465 → 0.575; HumanEval online 0.76 → 0.88                                                 | MMLU optimization 5.32 USD, inference 1.82 USD; HumanEval with optimization total 28.46 USD vs 1.61 USD, cannot be viewed as same-scope deployment cost        | ● ● ● |
| <b>AvaTaR</b><br>(S. Wu et al., 2024)              | NeurIPS 2024                                  | Instructions and action sequences; task-direct action program reuse                      | Knowledge base retrieval, multi-tool QA; training then testing official and human held-out queries | claude-3-opus STaRK Amazon/MAG/Prime top-1 hit rates 49.87 / 44.36 / 18.44 vs Reflexion 42.79 / 40.71 / 14.28                                    | Time and cost not measured; timeout-triggered rewrite is mechanism only, threshold not given                                                                   | ○ ○ ● |
| <b>DSPy</b><br>(Khattab et al., 2024)              | ICLR 2024                                     | Select prompts from successful trajectories and ensemble; fine-tuning branch not counted | GSM8K, HotPotQA; held-out test present, some report dev set only                                   | GPT-3.5 GSM8K uncompiled vanilla test 25.2 vs CoT bootstrap+ensemble 81.6; architecture and prompt change together, not single factor            | Compilation needs minutes to tens of minutes per author estimate; deployment tokens, time, cost not measured                                                   | ○ ● ● |
| <b>GEPA</b><br>(Agrawal et al., 2026)              | ICLR 2026                                     | Module prompts only, model frozen                                                        | QA, instructions, privacy rewriting, math; train validation test separated                         | Qwen3 8B AIME-2025: 32.00 vs MIPROv2 20.00, GRPO 38.00; overall advantage ≠ every task wins                                                      | Per-task optimization rollout total 6871 / 3593 / 7051 / 2426 / 1839 / 1839; GRPO per item 24000; not model calls, time or cost                                | ○ ● ● |
| <b>ACE</b><br>(Q. Zhang et al., 2026)              | ICLR 2026                                     | Text context entries, model frozen                                                       | AppWorld, finance; offline held-out and online task stream tested separately                       | DeepSeek-V3.1 AppWorld: offline labeled 59.4, unlabeled 57.2, baseline 42.4; no warm-up online 56.1; FiNER unlabeled online 67.3 < baseline 70.7 | AppWorld offline adaptation 9517 s vs GEPA 53898 s; online FiNER 2.9 vs DC 17.7 dollars; adaptation cost metric; statistical scope has unexplained differences | ● ● ● |
| <b>ESPO</b><br>(L. Liu et al., 2026)               | EMNLP 2026 Findings (listed in the programme) | Natural-language module prompt                                                           | Seven text benchmark, each 500 held-out; Claude Sonnet 4.5                                         | Main optimize single average 74.67% vs GEPA 70.91%; prompt 1,004 vs 1,878 character                                                              | PUPA optimize ~3.4M tokens, 2h 13m; no deployment acceleration measured                                                                                        | ○ ● ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A03 · Experience memory (1/4)

| WORK                                                       | VENUE     | WHAT IT CHANGES                                                                              | WHERE IT WAS TESTED                                                                           | EFFECT, WITH CONDITIONS                                                                                                                                     | TIME, MONEY, LEARNING COST                                                                                                                                     | MARKS |
|------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Dynamic Cheatsheet</b><br>(Suzgun et al., 2026)         | EACL 2026 | Online-maintained text, strategies and code snippets                                         | Math, science QA; accumulated in single test stream                                           | GPT-4o Game of 24: baseline 10%, fixed-prompt zero memory 19%, retrieved organized memory 99%; Claude 3.5 Sonnet baseline 12% → 14%                         | AIME 2024 Claude: 370 / 494 / 1035 / 1831 tokens; memory version higher; token types and management overhead range unclear; time and cost not measured         | ● ○ ● |
| <b>ACE</b><br>(Q. Zhang et al., 2026)                      | ICLR 2026 | Text context entries, model frozen                                                           | AppWorld, finance; offline held-out and online task stream tested separately                  | DeepSeek-V3.1 AppWorld: offline labeled 59.4, unlabeled 57.2, baseline 42.4; no warm-up online 56.1; FiNER unlabeled online 67.3 < baseline 70.7            | AppWorld offline adaptation 9517 s vs GEPA 53898 s; online FiNER 2.9 vs DC 17.7 dollars; adaptation cost metric; statistical scope has unexplained differences | ● ● ● |
| <b>Agent Workflow Memory (AWM)</b><br>(Wang et al., 2025a) | ICML 2025 | Infer text workflows from trajectories; action encapsulation has variants                    | WebArena online; Mind2Web cross-task, site, domain                                            | GPT-4-0613 WebArena: 35.5 vs BrowserGym 23.5 success rate; compared input contains HTML, not strict same interface; Mind2Web full-task success rate 0.7—4.8 | WebArena average steps 5.9 vs author rerun with accessibility-tree-only BrowserGym 7.9; step comparison not same as 23.5 success-rate row                      | ● ● ● |
| <b>ExpeL</b><br>(Zhao et al., 2024)                        | AAAI 2024 | Suggestion list and retrieved successful trajectories                                        | HotpotQA, ALFWorld, WebShop; train and eval tasks separated                                   | gpt-3.5-turbo-0613 ALFWorld single-attempt success 59% vs Reflexion after third retry 54%; different test budget                                            | Training retry limit 3 only; extraction calls, tokens, time or cost not measured                                                                               | ○ ○ ● |
| <b>MemRL</b><br>(S. Zhang et al., 2026)                    | arXiv     | Experience repository and retrieval value based on reward                                    | Code, OS, database, ALFWorld, HLE; repeated training vs frozen memory with held-out separated | Held-out average success rate vs MemP +2.8 pp; ALFWorld +5.8, OS +2.6; different baseline foundation models                                                 | HLE Gemini-3-pro average ~32K tokens per question across 10 epochs; whether measured per attempt unclear; no comparable MemP figures                           | ● ● ● |
| <b>MemQ</b><br>(Liao et al., 2026)                         | arXiv     | External experience value and generated relevance graph                                      | Multimodal QA, code, tools, OS; freeze best round memory for held-out                         | Gemma-4-E4B-it LiveCodeBench: 51.43±2.86 vs RAG 50.48±1.65, MemRL 45.71±2.86; 3 seeds; best round selection not clarified                                   | Time, cost, tokens or calls not measured; 'faster learning' means per-round success rate                                                                       | ○ ○ ● |
| <b>AEL</b><br>(Xu et al., 2026)                            |           | Main config learns memory retrieval selection, reflection, rules; code evolution default off | Claude Haiku 4.5 multi-task allocation, Gemini 3.1 Flash-Lite work-order stream               | Multi-task allocation test Sharpe 2.13 +/- 0.47; EvoTool 1.37 +/- 1.74, but difference p=0.39; planning evolution -1.72                                     | Author estimate full run \$2.80, prediction ~\$2, reflection evolution ~\$0.80; latency not measured                                                           | ○ ● ● |
| <b>Metis</b><br>(Dai et al., 2026)                         | arXiv     | Environment facts, pitfalls, plans and callable code from repeated plans                     | AppWorld; offline train once, freeze memory then held-out test                                | GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8                                                                                | Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A03 · Experience memory (2/4)

| WORK                                                 | VENUE                   | WHAT IT CHANGES                                                                            | WHERE IT WAS TESTED                                                                                                 | EFFECT, WITH CONDITIONS                                                                                                             | TIME, MONEY, LEARNING COST                                                                                                                             | MARKS |
|------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Mem<sup>2</sup>Evolve</b><br>(Cheng et al., 2026) | ACL 2026                | Expert agents, executable tools and experience                                             | General assistant, math, QA, planning, embodied, shopping; main result test stream accumulation inferred as entries | GPT-5-chat ALFWorld 94.31 vs AFLOW 93.40, HotpotQA 60.80 same as AFLOW; cross-task memory analysis has separate held-out            | Tool average repair rounds 1.01 → 0.48; tokens, time cost not measured; rounds not convertible to them                                                 | ● ○ ● |
| <b>FORGE</b><br>(Bogdanov et al., 2026)              | ACM CAIS 2026           | Multiple individual rules or example memory, champion broadcasts entire                    | CybORG CAGE-2 fixed attacker; train then freeze evaluation, same scenario                                           | Gemini best example scheme reward -24.5 vs zero-shot -189.6, closer to zero is better; other attackers not tested                   | Gemini rule scheme per-individual adaptation 7.3M +/- 1.6M tokens vs Reflexion 11.4M +/- 0.5M; time and cost not measured                              | ○ ● ● |
| <b>G-Memory</b><br>(G. Zhang et al., 2025)           | NeurIPS 2025            | Multi-agent interaction, task and suggestion associative memory                            | QA, text embodied, planning; independent memory build vs held-out not clear                                         | Qwen-2.5-14b + MacNet ALFWorld 58.21% → 79.10%; run count not reported                                                              | AutoGen+PDDL relative to no memory 1.4×10^6 tokens more, score 10.32 higher; management cost not separated; time and cost not measured                 | ● ○ ● |
| <b>CTIM-Rover</b><br>(Lindenbauer et al., 2025)      | ACL 2025 REALM workshop | General and repo-level text suggestions, can include successful trajectory examples        | SWE-bench Verified, 401 train, 45 test, also Lite trajectories                                                      | GPT-4o o1 extraction: full 40% vs AutoCodeRover 42%; text memory only 31%; whether extra Lite trajectories overlap not clarified    | Average turns 11.16 → 9.62; final context 11414.02 → 17544.93 tokens, not cumulative tokens; time and cost not measured                                | ● ○ ● |
| <b>EvolveMem</b><br>(J. Liu et al., 2026)            | arXiv                   | Retrieval config, answer style and re-extracted memory for gaps                            | LoCoMo conversational QA, MemBench; main performance held-out independence not reported                             | GPT-4o LoCoMo F1 0.543 vs SimpleMem 0.432; evolution same question 30.5 → 54.3                                                      | Single LoCoMo sample full evolution 25-35 min; retrieval 15 ms, validation ~2-3 s additional; no end-to-end latency                                    | ● ● ● |
| <b>DecentMem</b><br>(Hao et al., 2026)               | arXiv                   | Per-agent local experience pool and exploration/reuse router                               | Math, code, BBH, ALFWorld; held-out and retry count not reported                                                    | Qwen3-4B + AgentNet average accuracy 53.69 vs G-Memory 43.37, no memory 35.20                                                       | Qwen3-8B BBH vs G-Memory token reduction 32% / 49% / 47%, unclear if judging included; no-memory still cheaper                                         | ● ○ ● |
| <b>EXG</b><br>(Jin et al., 2026)                     | arXiv                   | Failure-success experience graph, connecting failures to repair cases                      | Code and multi-hop QA; online stream and frozen graph 7:3 held-out                                                  | Qwen3-Coder-Flash HumanEval held-out: EXG-Reflexion 0.879 vs ExpeL 0.909; online has larger gains                                   | HumanEval baseline EXG: 1.20 vs Reflexion 1.83 model calls; LLM latency 3,259 vs 3,416 ms; total tokens 125,304 vs 117,378; cost-graph model not clear | ● ○ ● |
| <b>Meta-TTL</b><br>(Lou et al., 2026)                |                         | Cross-round action prompts for same task and 'how to change prompt' prompt learned offline | Jericho, WebArena-Lite, tau2-bench; new instances and new domains                                                   | Gemini-3-Flash action, GPT-5 adaptation; WebArena-Lite unseen domain W-AUC 0.766 vs TextGrad 0.728, 3 runs, not single success rate | Jericho per session 860.9 s vs Static 593.2 s; 204.0 vs 231.0 calls; 364k vs 196k tokens; offline ~\$30-70/benchmark                                   | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A03 · Experience memory (3/4)

| WORK                                          | VENUE                        | WHAT IT CHANGES                                                                   | WHERE IT WAS TESTED                                                                                                       | EFFECT, WITH CONDITIONS                                                                                                                              | TIME, MONEY, LEARNING COST                                                                                                                            | MARKS |
|-----------------------------------------------|------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>SEDM</b><br>(Xu et al., 2025)              | NeurIPS 2025 SEA workshop    | Injectable memory fragments with utility weight and source                        | LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear                                                        | GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34                                                                            | FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured                                                  | ● ○ ● |
| <b>Agent S</b><br>(Agashe et al., 2025)       | ICLR 2025                    | Frozen model, update full-task and sub-task text memory; interface layer fixed    | GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration                                  | GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out | Checked entry does not report steps, time, tokens, cost or build cost                                                                                 | ○ ○ ● |
| <b>SE-Agent</b><br>(Guo et al., 2025)         | NeurIPS 2025                 | Candidate trajectory pool for same task, rewriting, recombining and filtering     | SWE-bench Verified; per-question search, not cross-question held-out memory                                               | GPT-4o: Pass@1 40.4% vs SWE-Agent 22.4%, SWE-Search 32.6%; internal multi-candidate, not single trajectory                                           | Default 10 candidates; max API cost graph unit unclear; actual tokens, time cost not measured                                                         | ○ ○ ○ |
| <b>WebCoach</b><br>(Liu et al., 2025)         | ICLR 2026 MemAgents workshop | External trajectory summary, additional Coach injects suggestions to action model | WebVoyager 643 real-time web tasks; dynamic library accumulates same-stream same-task but excludes same-task-ID retrieval | Skywork-38B + Qwen3-8B Coach: 0.614 vs 0.473; Qwen-VL-7B main table 0.311 vs 0.328                                                                   | Skywork average per question 215 → 395 s, actions 10.7 → 10.2; time includes extra inference; no token cost measurement                               | ● ○ ● |
| <b>ReasoningBank</b><br>(Ouyang et al., 2026) | ICLR 2026                    | Text experience pool extracted from success and failure trajectories              | WebArena, SWE-bench; online single-task stream                                                                            | WebArena Gemini-2.5-flash success rate 40.5 → 48.8; Shopping judge accuracy 72.7%                                                                    | WebArena average ReAct steps 9.7 → 8.3; another table total tokens 50,847.4 → 53,054.5, baseline foundation not specified; time and cost not measured | ● ○ ● |
| <b>ReAP</b><br>(Azam et al., 2025)            | arXiv                        | Text reflection retrieved by task similarity                                      | WebArena, same-task redo vs similar new-task tested separately                                                            | AgentOccam gpt-4o-2024-08-06; 80% build, 20% test new-task setup; reflection vs no memory +11 points; old success task best only 84%                 | Same-task redo average steps 11.92 → 10.08; new task step reduction 29.1%; time cost not measured; build reports one trajectory per task only         | ● ● ● |
| <b>Memento</b><br>(Zhou et al., 2025)         | arXiv                        | Case library; parameter branch additionally trains two-layer MLP retrieval scorer | GAIA and research QA; GAIA validation set builds library then test set                                                    | GPT-4.1 planning, o3 execution: GAIA validation Pass@3 87.88%, test set using only validation-accumulated library 79.40%; different protocols        | Each level sample 10 questions; GAIA L1 input/output 26k/4.7k, L3 121k/9.8k tokens; not complete learning cost, no savings comparison                 | ● ○ ● |
| <b>MobileGPT</b><br>(Lee et al., 2024)        | ACM MobiCom 2024             | Per-app page-subtask graph and action sequence                                    | Mobile apps; cold start then warm start same task with parameter change                                                   | GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix                                                | Human-repair path warm start vs same prompt Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78         | ● ● ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ○ partly, ○ no (initial judgment, calc.; Appendix A1).

# A03 · Experience memory (4/4)

| WORK                                                                                             | VENUE                                                | WHAT IT CHANGES                                                           | WHERE IT WAS TESTED                                                        | EFFECT, WITH CONDITIONS                                                                            | TIME, MONEY, LEARNING COST                                             | MARKS |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------|
| <b>Grounding Agent Memory</b><br>(Suresh et al., 2026)                                           | arXiv                                                | Only modify post-task experience library, add environment read-only probe | CLBench 40 question, unreported structure change; gpt-5.4; five paired run | Pass 39% → 73%; executor per-question SQL query 8.8 → 4.7                                          | Full execution \$3.38 → \$1.68; no experience-build and probe-maintain | ● ○ ● |
| <b>Are Online Skill and Memory Modules Always Worth Their Tokens?</b><br>(Hajimiri et al., 2026) | EMNLP 2026 main conference (listed in the programme) | Control AWM, ASI, ReasoningBank vs plain executor budget config           | WebArena four domain, Gemini 3 Flash, three run                            | Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously | Token only; plain 15 step+pruning, enhance 10 step, near-equal budget  | ● ● ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A04 · Skills and tools (1/4)

| WORK                                       | VENUE     | WHAT IT CHANGES                                                                                                  | WHERE IT WAS TESTED                                                                                                 | EFFECT, WITH CONDITIONS                                                                                                                              | TIME, MONEY, LEARNING COST                                                                                                                                     | MARKS |
|--------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Metis</b><br>(Dai et al., 2026)         | arXiv     | Environment facts, pitfalls, plans and callable code from repeated plans                                         | AppWorld; offline train once, freeze memory then held-out test                                                      | GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8                                                                         | Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection | ● ○ ● |
| <b>Mem²Evolve</b><br>(Cheng et al., 2026)  | ACL 2026  | Expert agents, executable tools and experience                                                                   | General assistant, math, QA, planning, embodied, shopping; main result test stream accumulation inferred as entries | GPT-5-chat ALFWorld 94.31 vs AFLOW 93.40, HotpotQA 60.80 same as AFLOW; cross-task memory analysis has separate held-out                             | Tool average repair rounds 1.01 → 0.48; tokens, time cost not measured; rounds not convertible to them                                                         | ● ○ ● |
| <b>Agent S</b><br>(Agashe et al., 2025)    | ICLR 2025 | Frozen model, update full-task and sub-task text memory; interface layer fixed                                   | GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration                            | GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out | Checked entry does not report steps, time, tokens, cost or build cost                                                                                          | ○ ○ ● |
| <b>SkillWeaver</b><br>(Zheng et al., 2025) | arXiv     | Playwright function library from web exploration, includes usage notes and page prerequisites                    | GPT-4o WebArena; self-practice then test benchmark tasks                                                            | Success rate 22.6 → 29.8; online human-judged subset 40.2 → 56.2, Cars 11.1 → 11.1                                                                   | Per site 160 exploration iterations; checked entry does not report runtime and exploration time cost                                                           | ○ ● ● |
| <b>Voyager</b><br>(G. Wang et al., 2024)   | TMLR 2024 | Executable Minecraft skills, completed and failed task tables                                                    | GPT-4; MineDojo/Mineflayer; exploration vs new worlds no task separation                                            | New world each unseen task 3/3 success; ReAct, Reflexion, AutoGPT 0/3 within 50 prompting iterations                                                 | '15.3× faster' is prompt-iteration ratio to unlock wood tools; seconds, tokens, dollars checked entry not reported                                             | ● ○ ● |
| <b>MetaAgent</b><br>(Qian et al., 2025)    | arXiv     | Frozen model, accumulate text experience and raw tool result knowledge base; no explicit environment-rules model | QwQ-32B GAIA/WebWalkerQA/BrowseComp; pre-fill library with same-batch task simulation                               | WebWalkerQA 52.1 vs WebThinker-32B-RL 46.5; GAIA 47.6 below its 48.5; no held-out split                                                              | Checked entry does not report measured cost or time                                                                                                            | ○ ○ ○ |
| <b>SkillOpt</b><br>(Y. Yang et al., 2026)  | arXiv     | Deploy skill documentation; execution model and framework fixed                                                  | GPT/Qwen; QA, table, document, math, ALFWorld; independent test                                                     | GPT-5.5 direct dialogue average 58.8 → 82.3 vs no skill; best per-cell baseline average 76.9                                                         | Training tokens e.g. SpreadsheetBench 21.4M; cost per question inconsistent scope; deployment per-task usage and time not reported                             | ○ ● ● |
| <b>VASO</b><br>(Y. Yang et al., 2026)      | arXiv     | Logic skill rules, proposition mapping, plan templates                                                           | Jackal/PX4 planning and simulation; GPT-5-nano skill generation, GPT-4o-mini planning                               | Automated rules satisfy 96.8, task completion 86.5; LLM-Planner 90.3/89.3; new prompt safety 92–100%, completion 80–96%                              | Author reported under 20 minutes per skill; skill execution time and cost not reported                                                                         | ○ ● ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A04 · Skills and tools (2/4)

| WORK                                                            | VENUE                        | WHAT IT CHANGES                                                                                                   | WHERE IT WAS TESTED                                                                     | EFFECT, WITH CONDITIONS                                                                                                                        | TIME, MONEY, LEARNING COST                                                                                            | MARKS |
|-----------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------|
| <b>AgentOrchestra</b><br>(W. Zhang et al., 2025)                | arXiv                        | Prompts, tool/agent/memory/environment code and scheme version                                                    | GAIA, HLE, GPQA, AIME; mainly Gemini 3 Flash Preview, browse component different models | HLE 55.2 → 59.6; GAIA Validation 89.70 → 93.33; independent evolution split not stated                                                         | Author rough estimate simple task 30 seconds/5k tokens, complex 10 minutes/100k; no Vanilla/Evolved cost comparison   | ● ○ ● |
| <b>GenericAgent</b><br>(Liang et al., 2026)                     | arXiv                        | SOP, script, hierarchical memory; fixed tool interface                                                            | Multiple models; local task, web, repeated download and research tasks                  | Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%        | Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated | ● ● ● |
| <b>STELLA</b><br>(Jin et al., 2025)                             | arXiv                        | Biomedical tool pool, reasoning template                                                                          | Claude 4 Sonnet + Gemini 2.5 Pro; HLE biomedical, LAB-Bench subset                      | HLE 14% → 26%; LitQA ~52% → 63% with trial budget increase; clear evolution/held-out separation missing                                        | Only trials 1x to 9x reported; time/token/cost not reported                                                           | ○ ● ● |
| <b>DRAFT / From Exploration to Mastery</b><br>(Qu et al., 2025) | ICLR 2025                    | Rewrite natural-language tool documentation, do not train execution model                                         | GPT-4o writes docs, GPT-4o/mini/Llama-3-70B use; RestBench, ToolBench                   | GPT-4o TMDb correct-path rate 88.00 vs ReAct 71.00; tool path includes standard subsequence, not final-state check                             | Checked entry does not report run and improvement call total, tokens, time, cost                                      | ○ ○ ● |
| <b>CODESKILL</b><br>(Y. Li et al., 2026)                        | arXiv                        | Text skill library + independently-trained SFT/RL management model                                                | Frozen Qwen3.5-35B-A3B or GPT-5.4-mini; programming and environment repair              | Qwen execution policy average 29.32 → 40.35; skill library built from test-set prerun, excluding current-task source, not strict online stream | Average reasoning steps 44.12 → 35.99; success-task scope inconsistent; manager RL ~210 hours, SFT ~20 hours          | ● ● ● |
| <b>LATM</b><br>(Cai et al., 2024)                               | ICLR 2024 main conference    | Python tools + call examples + tool cache                                                                         | GPT-4 create; GPT-3.5 Turbo/GPT-4 use; logic, bracket, sorting, schedule held-out       | GPT-3.5 Turbo Dyck Language 20.4 → 92.2; GPT-4 Logical Deduction 88.8 → 86.6                                                                   | Only O(nc+C) parse cost and call-budget design; time/tokens/dollars not measured                                      | ○ ● ● |
| <b>Alita-G</b><br>(Qiu et al., 2025b)                           | arXiv                        | Successful-trajectory tools abstracted into MCP toolbox, retrieval select                                         | Claude-Sonnet-4 manage + GPT-4.1 browse; GAIA validation set build, retest              | GAIA pass@1 75.15% → 83.03%; PathVQA/HLE build-test relationship not explained                                                                 | GAIA average generated tokens 12,305 → 10,394; build cost and latency not measured                                    | ● ○ ○ |
| <b>OS-Copilot / FRIDAY</b><br>(Z. Wu et al., 2024)              | ICLR 2024 LLMagents workshop | Frozen model, update program tool library; semantic memory is design component, experiment mainly validates tools | GAIA dev set learn, private test; self-created table tasks then SheetCopilot-20 test    | GPT-4 SheetCopilot-20 Pass@1 60% vs no learning 0%; GAIA Level 3 both 6.12%                                                                    | GAIA Level 1 table average time 118 vs 124 s; Level 2 221 vs 239 s; no-learning cost                                  | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A04 · Skills and tools (3/4)

| WORK                                          | VENUE                     | WHAT IT CHANGES                                                                      | WHERE IT WAS TESTED                                                                                        | EFFECT, WITH CONDITIONS                                                                                                                         | TIME, MONEY, LEARNING COST                                                                                                          | MARKS |
|-----------------------------------------------|---------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>OpenAgent</b><br>(Lyu et al., 2025)        | ACL 2025 main conference  | GitHub repo integration, Docker environment, experience summary                      | GPT-4, GPT-3.5; OpenAct; rerun experience experiment separate                                              | GPT-4 51-question environment/experience reuse 47.0 → 82.3; main table first-time task 47.3 vs ReAct 24.6, average scope not fully recomputable | Time/tokens/dollars checked entry not reported                                                                                      | ○ ○ ● |
| <b>OpenSkill</b><br>(Yan et al., 2026)        |                           | Per-target-task skill script + substitute test                                       | Opus 4.6/Claude Code, GPT 5.2/Codex; SkillsBench; same-task skill build then rescore                       | SkillsBench no skill → OpenSkill 25.5% → 43.6%, 25.0% → 42.1%; 5 evals per skill                                                                | Opus average run 465.0 → 845.4 s; skill build ~749K tokens/39 min; per-stage includes estimates, totals not fully compatible        | ● ● ● |
| <b>CoEvoSkills</b><br>(H. Zhang et al., 2026) | COLM 2026 main conference | Skill documentation, code and co-improved substitute test                            | Opus 4.6/Claude Code, GPT-5.2/Codex; SkillsBench same-question improvement and test                        | Opus pass rate 30.6% → 71.1%, manual skill 53.5%; not held-out generalization                                                                   | Average 4.1 validation cycle, 2.4 ground-truth round; seconds/tokens/dollars not measured                                           | ○ ● ● |
| <b>AgentDistill</b><br>(Qiu et al., 2025a)    | arXiv                     | Extract and merge MCP toolbox from frozen teacher trajectories for frozen student    | GPT-3.5, Qwen3-8B, LLaMA3.1-8B; Game of 24, medical VQA; held-out relationship not stated                  | GPT-3.5 Game of 24 accuracy 34.3±3.2 → 82.7±0.6; build 100 validation examples                                                                  | Only tool-call proportion reported; call count, time, tokens, dollars not measured                                                  | ● ○ ● |
| <b>CRAFT</b><br>(Yuan et al., 2024)           | ICLR 2024 main conference | Abstract and retrieve Python functions from solutions                                | GPT-4 build tools, GPT-3.5 Turbo main execution; VQA, TabMWP, MATH algebra; train source and eval separate | GPT-3.5 TabMWP 80.6 → 88.4, but same toolbox BM25 is 89.2; GQA SAcc 35.0 → 45.4                                                                 | Tool build author estimate ~\$2,500; runtime cost not measured                                                                      | ○ ● ● |
| <b>ActionEngine</b><br>(Zhong et al., 2026)   | arXiv                     | State-action graph, locator and descriptor, online general-purpose repair write-back | WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off                               | 91.2%; no warm-up graph 73.1%                                                                                                                   | 27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39-101 tasks and missing warm-up | ● ● ● |
| <b>WALT</b><br>(Prabhu et al., 2026)          | ICLR 2026 main conference | New action script, parameter constraint and tool description                         | WebArena, VisualWebArena; GPT-5 plan, GPT-5-mini execute                                                   | Average success 50.1% / 52.9%; same structure no-tool control best +30.7% relative, includes parser and verifier co-change                      | Steps best improve 1.4x; offline tokens, dollars, time not reported                                                                 | ● ● ● |
| <b>SpeedRunner</b><br>(Huang et al., 2026)    | arXiv                     | Frozen-model executable skill library, add/remove/modify functions                   | ScienceWorld, BabyAI, Crafter; gpt-5.4-mini; 200 online task, 30 held-out, 3 seed                          | BabyAI from ReAct ~67% to near-perfect; Crafter all-method average achievement unlock <30%                                                      | BabyAI usage ~one eighth; output-tokens vs dollars graph scope conflict, learning total cost not reported                           | ● ● ● |
| <b>ASI</b><br>(Wang et al., 2025b)            | COLM 2025 main conference | Python skill add to action library                                                   | WebArena online stream, no separate train set; claude-3.5-sonnet                                           | 40.4% vs plain Claude 32.7%; average action 5.0 vs 5.6, one skill = one action                                                                  | Learn tokens, dollars, time not reported; step scope not equal to raw operations                                                    | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A04 · Skills and tools (4/4)

| WORK                                                                                             | VENUE                                                | WHAT IT CHANGES                                                                                | WHERE IT WAS TESTED                                                                                             | EFFECT, WITH CONDITIONS                                                                            | TIME, MONEY, LEARNING COST                                                                                             | MARKS |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------|
| <b>AppAgentX</b><br>(Jiang et al., 2025)                                                         | arXiv                                                | Page/element relation memory and high-level action                                             | AppAgent 50 task/9 app; GPT-4o; five per-question                                                               | With chain memory then add action: success 70.8% → 71.4%; steps 9.1 → 5.7                          | Per-task tokens 9.26k → 4.94k; per-step time 23 → 16 s, successful-task-only, build cost unclear                       | ● ○ ● |
| <b>TraceCompiler</b><br>(Yadouni et al., 2026)                                                   | arXiv                                                | Data-dependency executable workflow                                                            | AppWorld one Venmo intent, three instance; author hand-translates execution                                     | Same instance tool calls 34 → 11; hold-one-instance 15 of 21 state test pass                       | Tool call only; compile cost and net save not tested                                                                   | ● ○ ● |
| <b>EchoPath</b><br>(Zhao et al., 2026)                                                           | arXiv                                                | Parameterized GUI trajectory memory, precondition and lifecycle                                | OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5-medium | 91.2% vs Synapse 91.8%                                                                             | Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only | ● ● ○ |
| <b>SKILL.nb</b><br>(Hattami et al., 2026)                                                        | ICML 2026 FAGEN workshop                             | Text step, code, pre/post check and version library, frozen model                              | WebArena-Verified 812 task, gpt-5.3-codex, shared framework different reasoning intensity                       | Single round 53.7%, +3.9 pp next; three perturbation reruns consistency 91.7%                      | Internal maintain/update 36k tokens per success, execution included unclear; no dollars or time                        | ● ● ● |
| <b>Are Online Skill and Memory Modules Always Worth Their Tokens?</b><br>(Hajimiri et al., 2026) | EMNLP 2026 main conference (listed in the programme) | Control AWM, ASI, ReasoningBank vs plain executor budget config                                | WebArena four domain, Gemini 3 Flash, three run                                                                 | Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously | Token only; plain 15 step+pruning, enhance 10 step, near-equal budget                                                  | ● ● ● |
| <b>ClawTrace / CostCraft</b><br>(Yuan et al., 2026)                                              | ACM CAIS 2026 Agent Skills workshop, non-archival    | Trajectory-summary exported skill document, frozen model                                       | SpreadsheetBench 30 held-out, OpenClaw + gpt-5.4; also migrate SkillsBench                                      | Remove delete patch after two-type sum degrade 5 → 14; migrate post cost median not decrease       | Execution cost with cache; SkillsBench median \$0.143 vs \$0.144; extract cost not reported                            | ● ○ ● |
| <b>AXIS</b><br>(Lu et al., 2025)                                                                 | ACL 2025 main conference                             | Translate UI trajectory into API-code skill                                                    | Microsoft Word, 50-task feasibility; location entry incomplete learn/test split list                            | Relative UFO: 84.0% vs 52.0%, 2.0 vs 3.2 step                                                      | 29.9 s vs 59.5 s; backend \$0.2 vs \$0.4; full learn cost location entry not reported                                  | ● ○ ● |
| <b>Skim</b><br>(Wong et al., 2026)                                                               | eligibility record: EuroSys 2027                     | URL template, answer structure and low-cost execution path                                     | WebVoyager/WebShop, WebVoyager/AgentOccam/BrowserUse framework                                                  | Cost median 1.9x lower, latency -33.4%, reported accuracy no decrease                              | Per-task dollars and end-to-end time, include fail and retry; offline amortize location entry not reported             | ● ○ ● |
| <b>Space</b><br>(Y. Yang et al., 2026)                                                           | EMNLP 2026 Findings                                  | Supervise policy model output multi-action block using skill boundary, deploy no skill library | ALFWorld, ScienceWorld; location entry does not list per-setting model and run count                            | Success improve 7.0%-31.3%, decision round best -78.9%; percentage meaning not itemized            | Only decision round and training step given; no time or dollars saved                                                  | ● ● ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ○ partly, ○ no (initial judgment, calc.; Appendix A1).

# A05 · Whole harness (1/4)

| WORK                                             | VENUE                     | WHAT IT CHANGES                                                                              | WHERE IT WAS TESTED                                                                     | EFFECT, WITH CONDITIONS                                                                                                                      | TIME, MONEY, LEARNING COST                                                                                                                              | MARKS |
|--------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Trace / OptoPrime</b><br>(Cheng et al., 2024) | NeurIPS 2024              | Prompts, function code or text-representable parameters                                      | BBH held-out questions, controlled held-out initial conditions, new chess positions     | BBH with reasoning template overall 78.6 vs DSPy prompt optimization 71.6; optimizer GPT-4-0125-Preview, in-program model not clearly stated | Word Sorting vs TextGrad: total experiment time 82.5 vs 211.1 min, including optimization, validation and testing; not per-question deployment latency  | ○ ● ● |
| <b>GPTSwarm</b><br>(Zhuge et al., 2024)          | ICML 2024 main conference | Node prompts, inter-agent connection probability; not LLM weights                            | GPT-3.5/GPT-4; MMLU, Crosswords, HumanEval                                              | Crosswords same 20 questions optimization evaluation 0.465 → 0.575; HumanEval online 0.76 → 0.88                                             | MMLU optimization 5.32 USD, inference 1.82 USD; HumanEval with optimization total 28.46 USD vs 1.61 USD, cannot be viewed as same-scope deployment cost | ● ● ● |
| <b>AEL</b><br>(Xu et al., 2026)                  |                           | Main config learns memory retrieval selection, reflection, rules; code evolution default off | Claude Haiku 4.5 multi-task allocation, Gemini 3.1 Flash-Lite work-order stream         | Multi-task allocation test Sharpe 2.13 +/- 0.47; EvoTool 1.37 +/- 1.74, but difference p=0.39; planning evolution -1.72                      | Author estimate full run \$2.80, prediction ~\$2, reflection evolution ~\$0.80; latency not measured                                                    | ○ ● ● |
| <b>EvolveMem</b><br>(J. Liu et al., 2026)        | arXiv                     | Retrieval config, answer style and re-extracted memory for gaps                              | LoCoMo conversational QA, MemBench; main performance held-out independence not reported | GPT-4o LoCoMo F1 0.543 vs SimpleMem 0.432; evolution same question 30.5 → 54.3                                                               | Single LoCoMo sample full evolution 25-35 min; retrieval 15 ms, validation ~2-3 s additional; no end-to-end latency                                     | ● ● ● |
| <b>SEDM</b><br>(Xu et al., 2025)                 | NeurIPS 2025 SEA workshop | Injectable memory fragments with utility weight and source                                   | LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear                      | GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34                                                                    | FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured                                                    | ● ○ ● |
| <b>GenericAgent</b><br>(Liang et al., 2026)      | arXiv                     | SOP, script, hierarchical memory; fixed tool interface                                       | Multiple models; local task, web, repeated download and research tasks                  | Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%      | Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated                                   | ● ● ● |
| <b>Symbolic Learning</b><br>(Ou et al., 2025)    |                           | Prompts, tools, node flow; model weights fixed                                               | GPT-3.5/GPT-4; QA, math, code, composition; split not explained                         | GPT-4 MATH 60.7 vs Agents 56.0; standard benchmark disables tools, cannot prove tool creation effective                                      | Time and cost checked entry not reported                                                                                                                | ○ ○ ● |
| <b>ADAS</b><br>(Hu et al., 2025)                 | ICLR 2025 main conference | Fixed small framework forward() program                                                      | GPT-4o search, GPT-3.5 execution; ARC, QA reasoning held-out test                       | DROP F1 79.4 vs 65.8; MGSM 53.4 vs 39.0, each strongest hand-written baseline                                                                | One search + eval ~500 USD (ARC), 300 USD (reasoning); deployment per-question not measured                                                             | ○ ● ● |
| <b>MOSS</b><br>(Cai et al., 2026b)               | arXiv                     | OpenClaw code and tool-result mediation                                                      | DeepSeek V3.2; claweval same four-task failures and retest                              | Average grader score 0.2526 → 0.6100, 3 per-question; no held-out task                                                                       | Only iteration-1 convergence reported; time/tokens/dollars not measured                                                                                 | ○ ● ○ |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A05 · Whole harness (2/4)

| WORK                                                          | VENUE                     | WHAT IT CHANGES                                                                        | WHERE IT WAS TESTED                                                                                    | EFFECT, WITH CONDITIONS                                                                                                                          | TIME, MONEY, LEARNING COST                                                                                              | MARKS |
|---------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------|
| <b>RewardHarness</b><br>(Y. Zhang et al., 2026)               | COLM 2026 main conference | Reward skill and tool-specification text library, vision model frozen                  | Qwen2.5-VL-7B/Gemini-2.0-Flash; image editing ranking                                                  | Qwen average 45.7 vs bare model 30.3, trained EditReward 42.0; do not treat downstream image-model training as no-training evidence              | 77 iteration scale only reported; tokens/seconds/dollars not measured                                                   | ○ ● ● |
| <b>STOP</b><br>(Zelikman et al., 2024)                        | COLM 2024 main conference | Python improver self-improve by calling fixed model                                    | GPT-4, GPT-3.5, Mixtral; small algorithm problems; held-out and new-task transfer                      | GPT-4-selected improver transfer 3SAT 21.2% → 75.1%; GPT-3.5, Mixtral average degrade                                                            | LPN ~3 thousand GPT-4 calls per round each run; deployment time dollars not measured                                    | ○ ● ● |
| <b>Huxley-Gödel Machine / HGM</b><br>(Wang et al., 2026)      | ICLR 2026 main conference | Agent code + search-tree expand/eval schedule                                          | SWE-bench, Polyglot; SWE GPT-5 modify, GPT-5-mini test                                                 | Same 800-eval budget SWE-Verified-60 40% → 56.7%; filter overlap SWE Lite held-out 34.8% → 40.1%                                                 | Optimization allocated CPU-hours: SWE 517 vs DGM 1231; not per-question deployment latency                              | ○ ● ● |
| <b>Gödel Agent</b><br>(Yin et al., 2025)                      | ACL 2025 main conference  | Policy code and self-modifying program                                                 | GPT-4o optimize, GPT-3.5 bounded execution; QA/math held-out test                                      | Bounded MGSM 64.2±3.4 vs Meta Agent Search 53.4±3.5; free version calls stronger model incomparable                                              | Full evolution author estimate ~\$15; per-question execution cost and time not measured                                 | ○ ● ● |
| <b>JudgeFlow</b><br>(Ma et al., 2026)                         | arXiv                     | Workflow logic blocks and configuration                                                | gpt-4o-mini; math/code; train-test split present, candidate-scoring split not clear                    | MATH 58.5 vs MermaidFlow 55.4; MATH → GSM8K 92.89 vs AFlow 91.95                                                                                 | Single GSM8K round evaluation \$0.45, Judge \$0.01; deployment time and cost not measured                               | ○ ● ● |
| <b>Adaptive Auto-Harness</b><br>(Z. Liu et al., 2026)         | arXiv                     | Task-type branching prompts, tools, infrastructure                                     | Sonnet4.6 execute, Opus4.6 modify; prediction market, security challenge, event forecast online stream | CTF-Dojo 37.2 → 50.2; FutureX 31.0 → 47.3, but A-Evolve 47.5                                                                                     | FutureX average time 244.4 → 47.6 s; PolyBench 18.2 → 42.2 s; improver tokens not saved, evolution overhead not counted | ● ○ ● |
| <b>Continual Harness</b><br>(Karten et al., 2026)             | arXiv                     | Online modify prompts, sub-agents, skills, memory; separate weight-training experiment | Pokémon Red/Emerald continuous no reset; Gemini series                                                 | Emerald Pro fixed 24 hours: improve from start 100% milestone, \$130 median vs minimal framework 98%, \$215; Flash-Lite improve reverse decline  | Measured per-run API cost; main behavior cost is button-press, not model call; improver cost not separately reported    | ● ● ● |
| <b>Red Queen Gödel Machine / RQGM</b><br>(Iacob et al., 2026) | arXiv                     | Execution and evaluator code co-modify; search-score shell fixed                       | GPT-5.5; code, paper, proof                                                                            | Polyglot held-out 71.7% vs HGM-H 69.9%, 119 vs 116 questions; each domain each baseline once                                                     | Baseline-score search tokens 1.35x-1.72x fewer; deployment per-question cost and time not measured                      | ○ ● ● |
| <b>Group-Evolving Agents / GEA</b><br>(Weng et al., 2026)     | arXiv                     | Multiple parent agents share trajectories then modify framework code                   | SWE-bench/Polyglot; Haiku4.5 then Sonnet4.5, o1 reflection                                             | Report SWE 71.0%, Polyglot 88.3%, but latter inconsistent with claimed 50-question held-out denominator; not use as generalization-gain evidence | Full cycle estimate USD 13,000 / USD 1,500; no per-question post-evolution cost change                                  | ○ ● ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A05 · Whole harness (3/4)

| WORK                                                       | VENUE                                                                         | WHAT IT CHANGES                                                                  | WHERE IT WAS TESTED                                                             | EFFECT, WITH CONDITIONS                                                                                    | TIME, MONEY, LEARNING COST                                                                                                                | MARKS |
|------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Hyperagents / DGM-H</b><br>(Zhang et al., 2026b)        | arXiv                                                                         | Executor and improver shared Python program                                      | Code, paper review, math grading; robot reward design has separate PPO training | Transfer improver to math grading imp@50=0.630, initial 0.0; 200 iterations 0.640 vs 0.610 not significant | Self-modify ~33M tokens, paper review + robot eval 0.506M/iteration; total and breakdown not fully compatible                             | ○ ● ● |
| <b>SICA</b><br>(Robeyns et al., 2025)                      | ICLR 2025 Self-Improving Foundation Models Without Human Supervision workshop | Self-complete Python code, tools and oversight mechanism                         | Sonnet3.5(v2), o3-mini; fixed coding subset and synthetic-file task             | SWE peak 17% → 53%, final accuracy 0.51; same baseline for agent selection, not held-out                   | Cross-four-benchmark full-attempt per-question average: final \$1.91 → \$1.70, 130.2 → 114.5 s; tokens 0.24 → 0.30 M; full cycle ~\$7,000 | ● ● ○ |
| <b>Harness-R1</b><br>(Shao et al., 2026)                   | arXiv                                                                         | Frozen target-outside execution hooks; separate engineering model train SFT+GRPO | Qwen3.5-9B; WebShop, ALFWorld, DBBench                                          | Main table same-batch 44.3% → 53.6%; also generate from 10 failures, 1,270 held-out +8.9 +/- 1.5 pp        | GPU config and training data number, actual time/tokens/dollars not measured                                                              | ○ ● ● |
| <b>AgentDevel</b><br>(Zhang, 2026)                         | arXiv                                                                         | Prompts, code, tool wrappers one release candidate                               | Sonnet4.5/Claude Code; SWE, WebArena, StableToolBench                           | SWE Lite 11.0% → 22.0%, Verified 15.0% → 30.0%; protocol keeps test, table naming not clear                | StableToolBench 11 rounds only; time/tokens/dollars not measured                                                                          | ○ ● ● |
| <b>Darwin Gödel Machine / DGM</b><br>(Zhang et al., 2026a) | ICLR 2026 main conference                                                     | Frozen model outside Python tools and workflow; search process fixed             | Sonnet3.5(New) modify, SWE same model, Polyglot o3-mini execute                 | SWE subset 20.0% → 50.0%; Polyglot full 14.2% → 30.7%, full includes search items                          | 80 iterations, single SWE run ~2 weeks; post-improvement per-question cost/latency not measured                                           | ○ ● ● |
| <b>Live-SWE-agent</b><br>(Xia et al., 2025)                | arXiv                                                                         | Per-question script create; framework unchanged, discard tool after completion   | SWE-bench Verified/Pro; multiple backbone                                       | Gemini3Pro success 74.2% → 77.4%; 50-question GPT-5-Nano 44.0% → 14.0%                                     | Same Gemini average per-question cost \$0.46 → \$0.48; no offline evolution does not mean total cost down                                 | ● ● ● |
| <b>RSEA</b><br>(Nguyen et al., 2026a)                      | arXiv                                                                         | Policy, skill rules, step manual text; ReAct loop unchanged                      | Qwen2.5-7B ALFWorld; Qwen3-30B WebShop; independent test                        | ALFWorld 69.3% vs ReAct 64.6%, but below Dynamic Cheatsheet 70.7 and difference not significant            | Tau-bench 190 meta+selection calls total; call scope not clear; time/tokens not measured                                                  | ○ ● ● |
| <b>HarnessFix</b><br>(Chen et al., 2026)                   | arXiv                                                                         | Prompts, tool description, orchestration, config, adapter and validation script  | GAIA, SWE-bench Verified, AppWorld, Terminal-Bench 2.0 Verified; GPT-5 mini     | Initial system to patched completion +6.3 to +18.4 percentage points; mean of three runs                   | AppWorld offline patch 37.2 million tokens; execution-phase time/cost not reported                                                        | ○ ● ● |
| <b>StarHarness</b><br>(Esakkiraja et al., 2026)            | arXiv                                                                         | Stirrup prompts, tools, skills, context, validation and end logic                | ITBench / EnterpriseOps-Gym / AutomationBench, GPT-5.4                          | Held-out increment +31.7 / +15.1 / +29.3 pp                                                                | All-benchmark per-task inference cost estimate -17% / -53% / -29%; offline optimize cost not reported                                     | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A05 · Whole harness (4/4)

| WORK                                                   | VENUE     | WHAT IT CHANGES                                            | WHERE IT WAS TESTED                                                                                  | EFFECT, WITH CONDITIONS                                                                   | TIME, MONEY, LEARNING COST                                                                                                  | MARKS |
|--------------------------------------------------------|-----------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Growing Harness</b><br>(L. Li et al., 2026)         | arXiv     | Frozen model and tools, only modify shared executable code | BrowseComp-Plus, WebArena-Verified × gpt-oss-120b/20b, Qwen3.5-4B; 50 final task per setting, 3 runs | Five of six groups highest success, one group -0.7 pp; deployment model calls -76.0-91.8% | Deployment inference cost -74.4-98.6%, not evaluator/offline improver; no break-even                                        | ● ○ ● |
| <b>Sol-Pi</b><br>(H. Liu et al., 2026)                 | arXiv     | Model fixed, modify run-framework context/tool mechanisms  | EdgeBench, GPT-5.6 Sol; held-out count statement inconsistent                                        | Full config score 44.8 → 42.0, token stream -49.0%; Performance config 44.8 → 47.2, -6.1% | Measure input/cache/output token stream; not total-search tokens, dollars or time                                           | ● ○ ● |
| <b>Agent JIT Compilation</b><br>(Winston et al., 2026) | ICML 2026 | Reuse code plan on tools, execution dispatch               | Five app, GPT-4.1/Gemini-2.5-Flash/Gemini-2.5-Pro average                                            | Planner average 11.7 s vs Browser-Use 122.1 s; 10.4x speedup                              | End-to-end time includes planning; plan score only dimensionless agent rank; offline total cost location entry not reported | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A06 · Predicting what an action does

| WORK                                                                 | VENUE                    | WHAT IT CHANGES                                                                                             | WHERE IT WAS TESTED                                                                               | EFFECT, WITH CONDITIONS                                                                                                             | TIME, MONEY, LEARNING COST                                                                                                                                     | MARKS |
|----------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>WMA</b><br>(Chae et al., 2025)                                    | ICLR 2025                | Train page-change prediction and scoring model; task execution model frozen                                 | GPT-4o WebArena same-environment synthetic instruction learn; also Mind2Web train/test            | WebArena 16.6% vs no-action-selection 12.8%, still below tree-search 19.2%; Mind2Web cross-task AWM+WMA 25.4% vs AWM w/o EF 15.3%   | Per-instruction 140.3 s vs search 748.3 s; table API \$0.4 vs \$2.7; learn-collection cost not reported; training ~3 GPU hours total scope unclear             | ●●●   |
| <b>Curious Causality-Seeking Agents / MCG</b><br>(Zhao et al., 2025) | NeurIPS 2025             | Train context encoder, causal graph and change-prediction network; no language model                        | Chemical unseen state, Magnetic manipulator arm; CEM planning                                     | Fork n=2 predict 63.18 ± 13.94 vs FCDL 57.82 ± 9.90; full model not every ablation winner                                           | Only training rounds and param size tested; no time/GPU hours/intervention cost                                                                                | ○●●   |
| <b>WorldEvolver</b><br>(X. Zhang et al., 2026)                       | arXiv                    | All models frozen, only modify environment model context: real change record, semantic rule, predict filter | Word2World predict, AgentBoard plan; memory cross-attempt and cross-task persist, Gemma-4-26B-A4B | ALFWorld predict exact-match 52.88 vs 3.60; plan best-of-5 26.12% vs 23.88%, first-try 19.4 below 22.4                              | Predict eval average 1.48 s/change vs zero-shot 1.05 s; total 5.24 vs 3.72 GPU hours; not plan-task cost                                                       | ●●●   |
| <b>Metis</b><br>(Dai et al., 2026)                                   | arXiv                    | Environment facts, pitfalls, plans and callable code from repeated plans                                    | AppWorld; offline train once, freeze memory then held-out test                                    | GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8                                                        | Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection | ●○●   |
| <b>Meta-TTL</b><br>(Lou et al., 2026)                                |                          | Cross-round action prompts for same task and 'how to change prompt' prompt learned offline                  | Jericho, WebArena-Lite, tau2-bench; new instances and new domains                                 | Gemini-3-Flash action, GPT-5 adaptation; WebArena-Lite unseen domain W-AUC 0.766 vs TextGrad 0.728, 3 runs, not single success rate | Jericho per session 860.9 s vs Static 593.2 s; 204.0 vs 231.0 calls; 364k vs 196k tokens; offline ~\$30-70/benchmark                                           | ●○●   |
| <b>AppWorld</b><br>(Trivedi et al., 2024)                            | ACL 2024 main conference | Build resettable apps and tasks, state checker; no agent improvement                                        | API programming operation simulate app; Test-N and Test-C include unseen app                      | GPT4O ReAct TGC 48.8% / 30.2%; fixed agent benchmark score, not learning gain                                                       | Current-price estimate ReAct Test-N \$0.7/task; judge average under 0.6 seconds, only checker                                                                  | ●—●   |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A07 · Descriptions, facts and preconditions (1/2)

| WORK                                                            | VENUE                        | WHAT IT CHANGES                                                                                                  | WHERE IT WAS TESTED                                                                                                       | EFFECT, WITH CONDITIONS                                                                                                                              | TIME, MONEY, LEARNING COST                                                                                                                                     | MARKS |
|-----------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Metis</b><br>(Dai et al., 2026)                              | arXiv                        | Environment facts, pitfalls, plans and callable code from repeated plans                                         | AppWorld; offline train once, freeze memory then held-out test                                                            | GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8                                                                         | Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection | ● ○ ● |
| <b>CTIM-Rover</b><br>(Lindenbauer et al., 2025)                 | ACL 2025 REALM workshop      | General and repo-level text suggestions, can include successful trajectory examples                              | SWE-bench Verified, 401 train, 45 test, also Lite trajectories                                                            | GPT-4o o1 extraction: full 40% vs AutoCodeRover 42%; text memory only 31%; whether extra Lite trajectories overlap not clarified                     | Average turns 11.16 → 9.62; final context 11414.02 → 17544.93 tokens, not cumulative tokens; time and cost not measured                                        | ● ○ ● |
| <b>Meta-TTL</b><br>(Lou et al., 2026)                           |                              | Cross-round action prompts for same task and 'how to change prompt' prompt learned offline                       | Jericho, WebArena-Lite, tau2-bench; new instances and new domains                                                         | Gemini-3-Flash action, GPT-5 adaptation; WebArena-Lite unseen domain W-AUC 0.766 vs TextGrad 0.728, 3 runs, not single success rate                  | Jericho per session 860.9 s vs Static 593.2 s; 204.0 vs 231.0 calls; 364k vs 196k tokens; offline ~\$30-70/benchmark                                           | ● ○ ● |
| <b>Agent S</b><br>(Agashe et al., 2025)                         | ICLR 2025                    | Frozen model, update full-task and sub-task text memory; interface layer fixed                                   | GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration                                  | GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out | Checked entry does not report steps, time, tokens, cost or build cost                                                                                          | ○ ○ ● |
| <b>WebCoach</b><br>(Liu et al., 2025)                           | ICLR 2026 MemAgents workshop | External trajectory summary, additional Coach injects suggestions to action model                                | WebVoyager 643 real-time web tasks; dynamic library accumulates same-stream same-task but excludes same-task-ID retrieval | Skywork-38B + Qwen3-8B Coach: 0.614 vs 0.473; Qwen-VL-7B main table 0.311 vs 0.328                                                                   | Skywork average per question 215 → 395 s, actions 10.7 → 10.2; time includes extra inference; no token cost measurement                                        | ● ○ ● |
| <b>MetaAgent</b><br>(Qian et al., 2025)                         | arXiv                        | Frozen model, accumulate text experience and raw tool result knowledge base; no explicit environment-rules model | QwQ-32B GAIA/WebWalkerQA/BrowseComp; pre-fill library with same-batch task simulation                                     | WebWalkerQA 52.1 vs WebThinker-32B-RL 46.5; GAIA 47.6 below its 48.5; no held-out split                                                              | Checked entry does not report measured cost or time                                                                                                            | ○ ○ ○ |
| <b>GenericAgent</b><br>(Liang et al., 2026)                     | arXiv                        | SOP, script, hierarchical memory; fixed tool interface                                                           | Multiple models; local task, web, repeated download and research tasks                                                    | Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%              | Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated                                          | ● ● ● |
| <b>DRAFT / From Exploration to Mastery</b><br>(Qu et al., 2025) | ICLR 2025                    | Rewrite natural-language tool documentation, do not train execution model                                        | GPT-4o writes docs, GPT-4o/mini/Llama-3-70B use; RestBench, ToolBench                                                     | GPT-4o TMDb correct-path rate 88.00 vs ReAct 71.00; tool path includes standard subsequence, not final-state check                                   | Checked entry does not report run and improvement call total, tokens, time, cost                                                                               | ○ ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A07 · Descriptions, facts and preconditions (2/2)

| WORK                                                   | VENUE                    | WHAT IT CHANGES                                                                | WHERE IT WAS TESTED                                                                                             | EFFECT, WITH CONDITIONS                                                       | TIME, MONEY, LEARNING COST                                                                                             | MARKS |
|--------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------|
| <b>AutoDroid-V2</b><br>(Wen et al., 2025)              | ACM MobiSys 2025         | App documentation, synthetic script and local model weight                     | DroidTask 158 task/13 app; Llama3.1 8B fine-tune; per-author explore and synthesis not look test set            | 54.4% vs same fine-tune step-by-step AutoDroid 43.9%                          | Phone model inference 46.3 s vs 669.2 s; does not include UI execution; offline prep and fine-tune separate            | ● ● ● |
| <b>EchoPath</b><br>(Zhao et al., 2026)                 | arXiv                    | Parameterized GUI trajectory memory, precondition and lifecycle                | OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5-medium | 91.2% vs Synapse 91.8%                                                        | Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only | ● ● ○ |
| <b>Unbrowse / Shadow APIs</b><br>(Tham et al., 2026)   | arXiv                    | Shared interface graph, parameter structure, response format, auth description | 94 domain read, single machine, hot cache third pass, include only aligned-output fields                        | Average request time 950 ms vs Playwright 3,404 ms                            | Measure end-to-end request time; 20-new-domain cold discover median 8,200 ms, publish success 18/20; no task cost      | ● ● ● |
| <b>SKILL.nb</b><br>(Hattami et al., 2026)              | ICML 2026 FAGEN workshop | Text step, code, pre/post check and version library, frozen model              | WebArena-Verified 812 task, gpt-5.3-codex, shared framework different reasoning intensity                       | Single round 53.7%, +3.9 pp next; three perturbation reruns consistency 91.7% | Internal maintain/update 36k tokens per success, execution included unclear; no dollars or time                        | ● ● ● |
| <b>Grounding Agent Memory</b><br>(Suresh et al., 2026) | arXiv                    | Only modify post-task experience library, add environment read-only probe      | CLBench 40 question, unreported structure change; gpt-5.4; five paired run                                      | Pass 39% → 73%; executor per-question SQL query 8.8 → 4.7                     | Full execution \$3.38 → \$1.68; no experience-build and probe-maintain                                                 | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A08 · Page and place structure

| WORK                                              | VENUE                            | WHAT IT CHANGES                                                                                | WHERE IT WAS TESTED                                                                   | EFFECT, WITH CONDITIONS                                                                                                                         | TIME, MONEY, LEARNING COST                                                                                                                       | MARKS |
|---------------------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>CTIM-Rover</b><br>(Lindenbauer et al., 2025)   | ACL 2025 REALM workshop          | General and repo-level text suggestions, can include successful trajectory examples            | SWE-bench Verified, 401 train, 45 test, also Lite trajectories                        | GPT-4o o1 extraction: full 40% vs AutoCodeRover 42%; text memory only 31%; whether extra Lite trajectories overlap not clarified                | Average turns 11.16 → 9.62; final context 11414.02 → 17544.93 tokens, not cumulative tokens; time and cost not measured                          | ● ○ ● |
| <b>MobileGPT</b><br>(Lee et al., 2024)            | ACM MobiCom 2024                 | Per-app page-subtask graph and action sequence                                                 | Mobile apps; cold start then warm start same task with parameter change               | GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix                                           | Human-repair path warm start vs same prompt<br>Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78 | ● ● ● |
| <b>Continual Harness</b><br>(Karten et al., 2026) | arXiv                            | Online modify prompts, sub-agents, skills, memory; separate weight-training experiment         | Pokémon Red/Emerald continuous no reset; Gemini series                                | Emerald Pro fixed 24 hours: improve from start 100% milestone, \$130 median vs minimal framework 98%, \$215; Flash-Lite improve reverse decline | Measured per-run API cost; main behavior cost is button-press, not model call; improver cost not separately reported                             | ● ● ● |
| <b>MrSteve</b><br>(Park et al., 2025)             | ICLR 2025                        | Save location, event, time and access graph; additionally train VPT-Nav, not all models frozen | Minecraft pixel control, sparse resource and long-horizon navigation; Steve-1 execute | Same sequence A-B-A task repeat A step count reduce; long-horizon ~70 tasks vs baselines <10, exact values in graph only                        | Memory query 11.24 ms vs FIFO 438.46 ms, not task time; navigator ~23 single-GPU training hours                                                  | ● ● ● |
| <b>ActionEngine</b><br>(Zhong et al., 2026)       | arXiv                            | State-action graph, locator and descriptor, online general-purpose repair write-back           | WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off          | 91.2%; no warm-up graph 73.1%                                                                                                                   | 27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39-101 tasks and missing warm-up              | ● ● ● |
| <b>AppAgentX</b><br>(Jiang et al., 2025)          | arXiv                            | Page/element relation memory and high-level action                                             | AppAgent 50 task/9 app; GPT-4o; five per-question                                     | With chain memory then add action: success 70.8% → 71.4%; steps 9.1 → 5.7                                                                       | Per-task tokens 9.26k → 4.94k; per-step time 23 → 16 s, successful-task-only, build cost unclear                                                 | ● ○ ● |
| <b>Skim</b><br>(Wong et al., 2026)                | eligibility record: EuroSys 2027 | URL template, answer structure and low-cost execution path                                     | WebVoyager/WebShop, WebVoyager/AgentOccam/BrowserUse framework                        | Cost median 1.9x lower, latency -33.4%, reported accuracy no decrease                                                                           | Per-task dollars and end-to-end time, include fail and retry; offline amortize location entry not reported                                       | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A09 · Exploring and practising (1/2)

| WORK                                                               | VENUE                        | WHAT IT CHANGES                                                                                                   | WHERE IT WAS TESTED                                                                               | EFFECT, WITH CONDITIONS                                                                                                                              | TIME, MONEY, LEARNING COST                                                                                            | MARKS |
|--------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------|
| <b>Agent S</b><br>(Agashe et al., 2025)                            | ICLR 2025                    | Frozen model, update full-task and sub-task text memory; interface layer fixed                                    | GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration          | GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out | Checked entry does not report steps, time, tokens, cost or build cost                                                 | ○ ○ ● |
| <b>SkillWeaver</b><br>(Zheng et al., 2025)                         | arXiv                        | Playwright function library from web exploration, includes usage notes and page prerequisites                     | GPT-4o WebArena; self-practice then test benchmark tasks                                          | Success rate 22.6 → 29.8; online human-judged subset 40.2 → 56.2, Cars 11.1 → 11.1                                                                   | Per site 160 exploration iterations; checked entry does not report runtime and exploration time cost                  | ○ ● ● |
| <b>Voyager</b><br>(G. Wang et al., 2024)                           | TMLR 2024                    | Executable Minecraft skills, completed and failed task tables                                                     | GPT-4; MineDojo/Mineflayer; exploration vs new worlds no task separation                          | New world each unseen task 3/3 success; ReAct, Reflexion, AutoGPT 0/3 within 50 prompting iterations                                                 | '15.3× faster' is prompt-iteration ratio to unlock wood tools; seconds, tokens, dollars checked entry not reported    | ● ○ ● |
| <b>GenericAgent</b><br>(Liang et al., 2026)                        | arXiv                        | SOP, script, hierarchical memory; fixed tool interface                                                            | Multiple models; local task, web, repeated download and research tasks                            | Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%              | Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated | ● ● ● |
| <b>OS-Copilot / FRIDAY</b><br>(Z. Wu et al., 2024)                 | ICLR 2024 LLMAgents workshop | Frozen model, update program tool library; semantic memory is design component, experiment mainly validates tools | GAIA dev set learn, private test; self-created table tasks then SheetCopilot-20 test              | GPT-4 SheetCopilot-20 Pass@1 60% vs no learning 0%; GAIA Level 3 both 6.12%                                                                          | GAIA Level 1 table average time 118 vs 124 s; Level 2 221 vs 239 s; no-learning cost                                  | ● ○ ● |
| <b>Planning to Explore / CovQValue</b><br>(Amayuelas et al., 2026) | arXiv                        | Frozen model, store code-branch coverage graph and test history, change test-candidate selection                  | RepoExploreBench/TestGenEval Lite; each program independent empty library, no cross-task held-out | Gemini 3 Flash TestGenEval Lite N=24 execution average branch 108.9 vs Greedy 61.5; coverage increase but more test fail                             | Execution count equal, model calls not equal; no total time/tokens/cost                                               | ○ ○ ○ |
| <b>SEAgent</b><br>(Sun et al., 2026)                               | ICML 2026 main conference    | Update software guide and curriculum, co-train actor and interface judge, beyond directional bound                | UI-TARS-7B OSWorld app subset; checked entry does not report held-out detail                      | Success 11.3% → 34.5%, latter first specialize then distill then generalist train; direct all-app train 30.6%, mean of three                         | 1k train iterations, 8 A100 80GB; no GPU hours, total time, tokens                                                    | ○ ● ● |
| <b>WALT</b><br>(Prabhu et al., 2026)                               | ICLR 2026 main conference    | New action script, parameter constraint and tool description                                                      | WebArena, VisualWebArena; GPT-5 plan, GPT-5-mini execute                                          | Average success 50.1% / 52.9%; same structure no-tool control best +30.7% relative, includes parser and verifier co-change                           | Steps best improve 1.4x; offline tokens, dollars, time not reported                                                   | ● ● ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A09 · Exploring and practising (2/2)

| WORK                             | VENUE                    | WHAT IT CHANGES                             | WHERE IT WAS TESTED                                                                  | EFFECT, WITH CONDITIONS                       | TIME, MONEY, LEARNING COST                                                            | MARKS |
|----------------------------------|--------------------------|---------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------|-------|
| <b>AXIS</b><br>(Lu et al., 2025) | ACL 2025 main conference | Translate UI trajectory into API-code skill | Microsoft Word, 50-task feasibility; location entry incomplete learn/test split list | Relative UFO: 84.0% vs 52.0%, 2.0 vs 3.2 step | 29.9 s vs 59.5 s; backend \$0.2 vs \$0.4; full learn cost location entry not reported | ● ○ ◐ |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ◐ partly, ○ no (initial judgment, calc.; Appendix A1).

# A10 · The works closest to ours (1/4)

| WORK                                            | VENUE                     | WHAT IT CHANGES                                                                      | WHERE IT WAS TESTED                                                                                  | EFFECT, WITH CONDITIONS                                                                                                    | TIME, MONEY, LEARNING COST                                                                                                          | MARKS |
|-------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>ActionEngine</b><br>(Zhong et al., 2026)     | arXiv                     | State-action graph, locator and descriptor, online general-purpose repair write-back | WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off                         | 91.2%; no warm-up graph 73.1%                                                                                              | 27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39-101 tasks and missing warm-up | ● ● ● |
| <b>WALT</b><br>(Prabhu et al., 2026)            | ICLR 2026 main conference | New action script, parameter constraint and tool description                         | WebArena, VisualWebArena; GPT-5 plan, GPT-5-mini execute                                             | Average success 50.1% / 52.9%; same structure no-tool control best +30.7% relative, includes parser and verifier co-change | Steps best improve 1.4x; offline tokens, dollars, time not reported                                                                 | ● ● ● |
| <b>SpeedRunner</b><br>(Huang et al., 2026)      | arXiv                     | Frozen-model executable skill library, add/remove/modify functions                   | ScienceWorld, BabyAI, Crafter; gpt-5.4-mini; 200 online task, 30 held-out, 3 seed                    | BabyAI from ReAct ~67% to near-perfect; Crafter all-method average achievement unlock <30%                                 | BabyAI usage ~one eighth; output-tokens vs dollars graph scope conflict, learning total cost not reported                           | ● ● ● |
| <b>ASI</b><br>(Wang et al., 2025b)              | COLM 2025 main conference | Python skill add to action library                                                   | WebArena online stream, no separate train set; claude-3.5-sonnet                                     | 40.4% vs plain Claude 32.7%; average action 5.0 vs 5.6, one skill = one action                                             | Learn tokens, dollars, time not reported; step scope not equal to raw operations                                                    | ● ○ ● |
| <b>HarnessFix</b><br>(Chen et al., 2026)        | arXiv                     | Prompts, tool description, orchestration, config, adapter and validation script      | GAIA, SWE-bench Verified, AppWorld, Terminal-Bench 2.0 Verified; GPT-5 mini                          | Initial system to patched completion +6.3 to +18.4 percentage points; mean of three runs                                   | AppWorld offline patch 37.2 million tokens; execution-phase time/cost not reported                                                  | ○ ● ● |
| <b>StarHarness</b><br>(Esakkiraja et al., 2026) | arXiv                     | Stirrup prompts, tools, skills, context, validation and end logic                    | ITBench / EnterpriseOps-Gym / AutomationBench, GPT-5.4                                               | Held-out increment +31.7 / +15.1 / +29.3 pp                                                                                | All-benchmark per-task inference cost estimate -17% / -53% / -29%; offline optimize cost not reported                               | ● ○ ● |
| <b>AutoDroid-V2</b><br>(Wen et al., 2025)       | ACM MobiSys 2025          | App documentation, synthetic script and local model weight                           | DroidTask 158 task/13 app; Llama3.1 8B fine-tune; per-author explore and synthesis not look test set | 54.4% vs same fine-tune step-by-step AutoDroid 43.9%                                                                       | Phone model inference 46.3 s vs 669.2 s; does not include UI execution; offline prep and fine-tune separate                         | ● ● ● |
| <b>AppAgentX</b><br>(Jiang et al., 2025)        | arXiv                     | Page/element relation memory and high-level action                                   | AppAgent 50 task/9 app; GPT-4o; five per-question                                                    | With chain memory then add action: success 70.8% → 71.4%; steps 9.1 → 5.7                                                  | Per-task tokens 9.26k → 4.94k; per-step time 23 → 16 s, successful-task-only, build cost unclear                                    | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A10 · The works closest to ours (2/4)

| WORK                                                                                             | VENUE                                                | WHAT IT CHANGES                                                                | WHERE IT WAS TESTED                                                                                             | EFFECT, WITH CONDITIONS                                                                            | TIME, MONEY, LEARNING COST                                                                                             | MARKS |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Growing Harness</b><br>(L. Li et al., 2026)                                                   | arXiv                                                | Frozen model and tools, only modify shared executable code                     | BrowseComp-Plus, WebArena-Verified × gpt-oss-120b/20b, Qwen3.5-4B; 50 final task per setting, 3 runs            | Five of six groups highest success, one group -0.7 pp; deployment model calls -76.0-91.8%          | Deployment inference cost -74.4-98.6%, not evaluator/offline improver; no break-even                                   | ● ○ ● |
| <b>TraceCompiler</b><br>(Yadouni et al., 2026)                                                   | arXiv                                                | Data-dependency executable workflow                                            | AppWorld one Venmo intent, three instance; author hand-translates execution                                     | Same instance tool calls 34 → 11; hold-one-instance 15 of 21 state test pass                       | Tool call only; compile cost and net save not tested                                                                   | ● ○ ● |
| <b>EchoPath</b><br>(Zhao et al., 2026)                                                           | arXiv                                                | Parameterized GUI trajectory memory, precondition and lifecycle                | OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5-medium | 91.2% vs Synapse 91.8%                                                                             | Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only | ● ● ○ |
| <b>Unbrowse / Shadow APIs</b><br>(Tham et al., 2026)                                             | arXiv                                                | Shared interface graph, parameter structure, response format, auth description | 94 domain read, single machine, hot cache third pass, include only aligned-output fields                        | Average request time 950 ms vs Playwright 3,404 ms                                                 | Measure end-to-end request time; 20-new-domain cold discover median 8,200 ms, publish success 18/20; no task cost      | ● ● ● |
| <b>SKILL.nb</b><br>(Hattami et al., 2026)                                                        | ICML 2026 FAGEN workshop                             | Text step, code, pre/post check and version library, frozen model              | WebArena-Verified 812 task, gpt-5.3-codex, shared framework different reasoning intensity                       | Single round 53.7%, +3.9 pp next; three perturbation reruns consistency 91.7%                      | Internal maintain/update 36k tokens per success, execution included unclear; no dollars or time                        | ● ● ● |
| <b>Grounding Agent Memory</b><br>(Suresh et al., 2026)                                           | arXiv                                                | Only modify post-task experience library, add environment read-only probe      | CLBench 40 question, unreported structure change; gpt-5.4; five paired run                                      | Pass 39% → 73%; executor per-question SQL query 8.8 → 4.7                                          | Full execution \$3.38 → \$1.68; no experience-build and probe-maintain                                                 | ● ○ ● |
| <b>Are Online Skill and Memory Modules Always Worth Their Tokens?</b><br>(Hajimiri et al., 2026) | EMNLP 2026 main conference (listed in the programme) | Control AWM, ASI, ReasoningBank vs plain executor budget config                | WebArena four domain, Gemini 3 Flash, three run                                                                 | Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously | Token only; plain 15 step+pruning, enhance 10 step, near-equal budget                                                  | ● ● ● |
| <b>ClawTrace / CostCraft</b><br>(Yuan et al., 2026)                                              | ACM CAIS 2026 Agent Skills workshop, non-archival    | Trajectory-summary exported skill document, frozen model                       | SpreadsheetBench 30 held-out, OpenClaw + gpt-5.4; also migrate SkillsBench                                      | Remove delete patch after two-type sum degrade 5 → 14; migrate post cost median not decrease       | Execution cost with cache; SkillsBench median \$0.143 vs \$0.144; extract cost not reported                            | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A10 · The works closest to ours (3/4)

| WORK                                                       | VENUE                                            | WHAT IT CHANGES                                                           | WHERE IT WAS TESTED                                                                  | EFFECT, WITH CONDITIONS                                                                                                                                     | TIME, MONEY, LEARNING COST                                                                                                                            | MARKS |
|------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>ESPO</b><br>(L. Liu et al., 2026)                       | EMNLP 2026<br>Findings (listed in the programme) | Natural-language module prompt                                            | Seven text benchmark, each 500 held-out; Claude Sonnet 4.5                           | Main optimize single average 74.67% vs GEPA 70.91%; prompt 1,004 vs 1,878 character                                                                         | PUPA optimize ~3.4M tokens, 2h 13m; no deployment acceleration measured                                                                               | ○ ● ● |
| <b>Sol-Pi</b><br>(H. Liu et al., 2026)                     | arXiv                                            | Model fixed, modify run-framework context/tool mechanisms                 | EdgeBench, GPT-5.6 Sol; held-out count statement inconsistent                        | Full config score 44.8 → 42.0, token stream -49.0%; Performance config 44.8 → 47.2, -6.1%                                                                   | Measure input/cache/output token stream; not total-search tokens, dollars or time                                                                     | ● ○ ● |
| <b>AXIS</b><br>(Lu et al., 2025)                           | ACL 2025 main conference                         | Translate UI trajectory into API-code skill                               | Microsoft Word, 50-task feasibility; location entry incomplete learn/test split list | Relative UFO: 84.0% vs 52.0%, 2.0 vs 3.2 step                                                                                                               | 29.9 s vs 59.5 s; backend \$0.2 vs \$0.4; full learn cost location entry not reported                                                                 | ● ○ ● |
| <b>Agent JIT Compilation</b><br>(Winston et al., 2026)     | ICML 2026                                        | Reuse code plan on tools, execution dispatch                              | Five app, GPT-4.1/Gemini-2.5-Flash/Gemini-2.5-Pro average                            | Planner average 11.7 s vs Browser-Use 122.1 s; 10.4x speedup                                                                                                | End-to-end time includes planning; plan score only dimensionless agent rank; offline total cost location entry not reported                           | ● ○ ● |
| <b>Skim</b><br>(Wong et al., 2026)                         | eligibility record: EuroSys 2027                 | URL template, answer structure and low-cost execution path                | WebVoyager/WebShop, WebVoyager/AgentOccam/BrowserUse framework                       | Cost median 1.9x lower, latency -33.4%, reported accuracy no decrease                                                                                       | Per-task dollars and end-to-end time, include fail and retry; offline amortize location entry not reported                                            | ● ○ ● |
| <b>Agent Workflow Memory (AWM)</b><br>(Wang et al., 2025a) | ICML 2025                                        | Infer text workflows from trajectories; action encapsulation has variants | WebArena online; Mind2Web cross-task, site, domain                                   | GPT-4-0613 WebArena: 35.5 vs BrowserGym 23.5 success rate; compared input contains HTML, not strict same interface; Mind2Web full-task success rate 0.7—4.8 | WebArena average steps 5.9 vs author rerun with accessibility-tree-only BrowserGym 7.9; step comparison not same as 23.5 success-rate row             | ● ● ● |
| <b>ReasoningBank</b><br>(Ouyang et al., 2026)              | ICLR 2026                                        | Text experience pool extracted from success and failure trajectories      | WebArena, SWE-bench; online single-task stream                                       | WebArena Gemini-2.5-flash success rate 40.5 → 48.8; Shopping judge accuracy 72.7%                                                                           | WebArena average ReAct steps 9.7 → 8.3; another table total tokens 50,847.4 → 53,054.5, baseline foundation not specified; time and cost not measured | ● ○ ● |
| <b>MobileGPT</b><br>(Lee et al., 2024)                     | ACM MobiCom 2024                                 | Per-app page-subtask graph and action sequence                            | Mobile apps; cold start then warm start same task with parameter change              | GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix                                                       | Human-repair path warm start vs same prompt Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78         | ● ● ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# A10 · The works closest to ours (4/4)

| WORK                                        | VENUE                     | WHAT IT CHANGES                                                                               | WHERE IT WAS TESTED                                                    | EFFECT, WITH CONDITIONS                                                                                                                 | TIME, MONEY, LEARNING COST                                                                                                                                     | MARKS |
|---------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>SkillWeaver</b><br>(Zheng et al., 2025)  | arXiv                     | Playwright function library from web exploration, includes usage notes and page prerequisites | GPT-4o WebArena; self-practice then test benchmark tasks               | Success rate 22.6 → 29.8; online human-judged subset 40.2 → 56.2, Cars 11.1 → 11.1                                                      | Per site 160 exploration iterations; checked entry does not report runtime and exploration time cost                                                           | ○ ● ● |
| <b>Metis</b><br>(Dai et al., 2026)          | arXiv                     | Environment facts, pitfalls, plans and callable code from repeated plans                      | AppWorld; offline train once, freeze memory then held-out test         | GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8                                                            | Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection | ● ○ ● |
| <b>GenericAgent</b><br>(Liang et al., 2026) | arXiv                     | SOP, script, hierarchical memory; fixed tool interface                                        | Multiple models; local task, web, repeated download and research tasks | Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80% | Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated                                          | ● ● ● |
| <b>Harness-R1</b><br>(Shao et al., 2026)    | arXiv                     | Frozen target-outside execution hooks; separate engineering model train SFT+GRPO              | Qwen3.5-9B; WebShop, ALFWorld, DBBench                                 | Main table same-batch 44.3% → 53.6%; also generate from 10 failures, 1,270 held-out +8.9 +/- 1.5 pp                                     | GPU config and training data number, actual time/tokens/dollars not measured                                                                                   | ○ ● ● |
| <b>SEDM</b><br>(Xu et al., 2025)            | NeurIPS 2025 SEA workshop | Injectable memory fragments with utility weight and source                                    | LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear     | GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34                                                               | FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured                                                           | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# D · Evidence behind the five difficulties (1/2)

| WORK                                                                                                        | VENUE                                                       | WHAT IT CHANGES                                                                            | WHERE IT WAS TESTED                                                                                                                | EFFECT, WITH CONDITIONS                                                                                                                       | TIME, MONEY, LEARNING COST                                                                                                                                  | MARKS |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>WebCoach</b><br>(Liu et al., 2025)                                                                       | ICLR 2026<br>MemAgents<br>workshop                          | External trajectory summary,<br>additional Coach injects suggestions<br>to action model    | WebVoyager 643 real-time web tasks;<br>dynamic library accumulates same-stream<br>same-task but excludes same-task-ID<br>retrieval | Skywork-38B + Qwen3-8B Coach: 0.614 vs 0.473;<br>Qwen-VL-7B main table 0.311 vs 0.328                                                         | Skywork average per question 215 → 395 s, actions<br>10.7 → 10.2; time includes extra inference; no token<br>cost measurement                               | ● ○ ● |
| <b>ReasoningBank</b><br>(Ouyang et al., 2026)                                                               | ICLR 2026                                                   | Text experience pool extracted from<br>success and failure trajectories                    | WebArena, SWE-bench; online single-task<br>stream                                                                                  | WebArena Gemini-2.5-flash success rate 40.5 →<br>48.8; Shopping judge accuracy 72.7%                                                          | WebArena average ReAct steps 9.7 → 8.3; another<br>table total tokens 50,847.4 → 53,054.5, baseline<br>foundation not specified; time and cost not measured | ● ○ ● |
| <b>GenericAgent</b><br>(Liang et al., 2026)                                                                 | arXiv                                                       | SOP, script, hierarchical memory;<br>fixed tool interface                                  | Multiple models; local task, web, repeated<br>download and research tasks                                                          | Same GPT-5.4 repeated download 102 seconds →<br>~66 seconds; not held-out; five long tasks success<br>100%, vs Claude Code 100%, OpenClaw 80% | Repeated download tokens 200,439 → 100,000; five<br>long tasks 220.8 s vs OpenClaw 183.1 s; evolution<br>cost not separated                                 | ● ● ● |
| <b>EchoPath</b><br>(Zhao et al., 2026)                                                                      | arXiv                                                       | Parameterized GUI trajectory<br>memory, precondition and lifecycle                         | OSWorld-Verified 159 task with<br>established memory; same-task second<br>try change resolution, codex-gpt-5.5-<br>medium          | 91.2% vs Synapse 91.8%                                                                                                                        | Median tokens 20,370 vs 586,386; time 127.5 s vs<br>315.7 s; first build median ~572k tokens/4.5 min<br>counted success-only                                | ● ● ○ |
| <b>MobileGPT</b><br>(Lee et al., 2024)                                                                      | ACM MobiCom<br>2024                                         | Per-app page-subtask graph and<br>action sequence                                          | Mobile apps; cold start then warm start<br>same task with parameter change                                                         | GPT-4-turbo; with human repair ablation warm start<br>98.75%; main table without human repair cannot mix                                      | Human-repair path warm start vs same prompt<br>Derive-only: latency -62.5%, cost -68.8%; offline<br>explore per app 10 to 15 minutes, total \$10.78         | ● ● ● |
| <b>ActionEngine</b><br>(Zhong et al., 2026)                                                                 | arXiv                                                       | State-action graph, locator and<br>descriptor, online general-purpose<br>repair write-back | WebArena 655 tasks, Claude Opus 4.6;<br>template warm-up then eval, patcher off                                                    | 91.2%; no warm-up graph 73.1%                                                                                                                 | 27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative<br>Claude Code; no explore/warm-up; break-even 39-<br>101 tasks and missing warm-up                  | ● ● ● |
| <b>Are Online Skill and<br/>Memory Modules Always<br/>Worth Their Tokens?</b><br>(Hajimiri et al.,<br>2026) | EMNLP 2026 main<br>conference (listed<br>in the programme)  | Control AWM, ASI, ReasoningBank vs<br>plain executor budget config                         | WebArena four domain, Gemini 3 Flash,<br>three run                                                                                 | Vanilla-IB success 44.78%, 73.6K total<br>tokens/question, that model four-config best<br>simultaneously                                      | Token only; plain 15 step+pruning, enhance 10 step,<br>near-equal budget                                                                                    | ● ● ● |
| <b>ClawTrace / CostCraft</b><br>(Yuan et al., 2026)                                                         | ACM CAIS 2026<br>Agent Skills<br>workshop, non-<br>archival | Trajectory-summary exported skill<br>document, frozen model                                | SpreadsheetBench 30 held-out, OpenClaw<br>+ gpt-5.4; also migrate SkillsBench                                                      | Remove delete patch after two-type sum degrade 5<br>→ 14; migrate post cost median not decrease                                               | Execution cost with cache; SkillsBench median<br>\$0.143 vs \$0.144; extract cost not reported                                                              | ● ○ ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

# D · Evidence behind the five difficulties (2/2)

| WORK                                                      | VENUE                                                                         | WHAT IT CHANGES                                                                 | WHERE IT WAS TESTED                                                         | EFFECT, WITH CONDITIONS                                                                                                                          | TIME, MONEY, LEARNING COST                                                                                                                | MARKS   |
|-----------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>SEDM</b><br>(Xu et al., 2025)                          | NeurIPS 2025 SEA workshop                                                     | Injectable memory fragments with utility weight and source                      | LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear          | GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34                                                                        | FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured                                      | ● ○ ●   |
| <b>HarnessFix</b><br>(Chen et al., 2026)                  | arXiv                                                                         | Prompts, tool description, orchestration, config, adapter and validation script | GAIA, SWE-bench Verified, AppWorld, Terminal-Bench 2.0 Verified; GPT-5 mini | Initial system to patched completion +6.3 to +18.4 percentage points; mean of three runs                                                         | AppWorld offline patch 37.2 million tokens; execution-phase time/cost not reported                                                        | ○ ● ● ● |
| <b>ESPO</b><br>(L. Liu et al., 2026)                      | EMNLP 2026 Findings (listed in the programme)                                 | Natural-language module prompt                                                  | Seven text benchmark, each 500 held-out; Claude Sonnet 4.5                  | Main optimize single average 74.67% vs GEPA 70.91%; prompt 1,004 vs 1,878 character                                                              | PUPA optimize ~3.4M tokens, 2h 13m; no deployment acceleration measured                                                                   | ○ ● ● ● |
| <b>GEPA</b><br>(Agrawal et al., 2026)                     | ICLR 2026                                                                     | Module prompts only, model frozen                                               | QA, instructions, privacy rewriting, math; train validation test separated  | Qwen3 8B AIME-2025: 32.00 vs MIPROv2 20.00, GRPO 38.00; overall advantage ≠ every task wins                                                      | Per-task optimization rollout total 6871 / 3593 / 7051 / 2426 / 1839 / 1839; GRPO per item 24000; not model calls, time or cost           | ○ ● ● ● |
| <b>SICA</b><br>(Robeyns et al., 2025)                     | ICLR 2025 Self-Improving Foundation Models Without Human Supervision workshop | Self-complete Python code, tools and oversight mechanism                        | Sonnet3.5(v2), o3-mini; fixed coding subset and synthetic-file task         | SWE peak 17% → 53%, final accuracy 0.51; same baseline for agent selection, not held-out                                                         | Cross-four-benchmark full-attempt per-question average: final \$1.91 → \$1.70, 130.2 → 114.5 s; tokens 0.24 → 0.30 M; full cycle ~\$7,000 | ● ● ● ○ |
| <b>ADAS</b><br>(Hu et al., 2025)                          | ICLR 2025 main conference                                                     | Fixed small framework forward() program                                         | GPT-4o search, GPT-3.5 execution; ARC, QA reasoning held-out test           | DROP F1 79.4 vs 65.8; MGSM 53.4 vs 39.0, each strongest hand-written baseline                                                                    | One search + eval ~500 USD (ARC), 300 USD (reasoning); deployment per-question not measured                                               | ○ ● ● ● |
| <b>Group-Evolving Agents / GEA</b><br>(Weng et al., 2026) | arXiv                                                                         | Multiple parent agents share trajectories then modify framework code            | SWE-bench/Polyglot; Haiku4.5 then Sonnet4.5, o1 reflection                  | Report SWE 71.0%, Polyglot 88.3%, but latter inconsistent with claimed 50-question held-out denominator; not use as generalization-gain evidence | Full cycle estimate USD 13,000 / USD 1,500; no per-question post-evolution cost change                                                    | ○ ● ● ● |

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).